Mechanics

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1 力學（Mechanics）

Unless otherwise stated, bodies, objects, and systems refer to insulated systems, positions are in inertial reference frames, t is time, and units are SI units.

I 簡史

- 150：托勒密（Ptolemy）提出地心模型（Geocentric model）。
- 1543：哥白尼（Nicolaus Copernicus）著「天體運行論（On the Revolutions of the Heavenly Spheres)」，發表日心說（Heliocentrism）。
- 1609：伽利略（Galileo Galilei）使用望遠鏡觀測行星。
- 1609：克卜勒（Johannes Kepler Kepler）著「新天文學（Astronomia nova)」，發表克卜勒第一及第二行星運動定律（Kepler's First and Second Laws of Planetary Motion）。
- 1619：克卜勒著「世界的和諧（Harmonices Mundi)」，發表克卜勒第三行星運動定律（Kepler's Third Law of Planetary Motion）。
- 1632：伽利略著「關於兩大世界體系的對話（Dialogue Concerning the Two Chief World Systems)」。
- 1634：伽利略提出慣性定律（Law of Inertia）。
- 1687：牛頓出版「自然哲學的數學原理（Philosophiae Naturalis Principia Mathematica）」，發表牛頓運動定律（Newton's Laws of Motion）與牛頓萬有引力定律（Newton's Law of Universal Gravitation）。

II 時刻與時間

i Time, moment, or instant (時刻)

At $t = k$ is called

- at the kth unit of time (在第 k 時間單位),
- at the beginning of the $(k + 1)$th unit of time (在第 $k + 1$ 時間單位初), or
- at the end of the kth unit of time (在第 k 時間單位末).

ii Time interval or duration (時間)

- $t = k - 1$ to $t = k$ is called during or within the kth unit of time (在第 k 時間單位內).
- $t = m$ to $t = m + k$ is called over the k-unit-of-time interval (since the mth unit of time) (在（自第 m 時間單位的）k 時間單位內).

III Motion of translation (平移運動)

i Vectors

- **Force F**：An interaction between system that satisfies the principle of superposition.
- **External force**: A force acting on a system from outside of it.
- **Resultant force**: The sum of all forces acting on a system.
- **Internal force**: A force between different subsystems in a system.
- **Contact Force (接觸力)**: A force that only exists when the two systems contact.
- **Non-contact force (非接觸力)/force at a distance (超距力)**: A force that can exist without the two systems to contact.
- **Tension (張力)**: Attraction contact forces within a system, which is balanced inside non-deformed materials.
- **Relative position**: The relative position of a point with respect to a given reference point (參考點).
- **Position r**: The position when an origin is chosen.
- **Displacement (位移)** $\mathbf{r}(t_2) - \mathbf{r}(t_1)$
- **(Instantaneous (瞬間/瞬時)) velocity(速度)** $\mathbf{v} = \dfrac{\mathrm{d}\mathbf{r}}{\mathrm{d}t}$
- **Average velocity** $\dfrac{\mathbf{r}(t_2) - \mathbf{r}(t_1)}{t_2 - t_1} = \dfrac{\int_{t_1}^{t_2} \mathbf{v}\,\mathrm{d}t}{t_2 - t_1}$
- **(Instantaneous) acceleration** $\mathbf{a} = \dfrac{\mathrm{d}\mathbf{v}}{\mathrm{d}t}$
- **(Instantaneous) jerk or jolt** $\dfrac{\mathrm{d}\mathbf{a}}{\mathrm{d}t}$
- **Average acceleration** $\dfrac{\mathbf{v}(t_2) - \mathbf{v}(t_1)}{t_2 - t_1} = \dfrac{\int_{t_1}^{t_2} \mathbf{a}\,\mathrm{d}t}{t_2 - t_1}$
- **(Linear) momentum (動量)** $\mathbf{p} = m\mathbf{v}$
- **Impulse (衝量)** $\mathbf{J} = \mathbf{p}(t_2) - \mathbf{p}(t_1)$

ii Scalars

- **Distance** $\left|\mathbf{r}_2 - \mathbf{r}_1\right|$
- **(Inertial (慣性)) mass (質量)** m: For a system when the force is applied to, the ratio of the force to the acceleration of the system.
- **(Instantaneous) speed (速率)** $|\mathbf{v}|$

- **Path length (路徑長)** $\int_{t_1}^{t_2} |\mathbf{v}| \, \mathrm{d}t$
- **Average speed** $\dfrac{\int_{t_1}^{t_2} |\mathbf{v}| \, \mathrm{d}t}{t_2 - t_1}$

iii Newton's first law of motion (牛頓第一運動定律)/Law of inertia (慣性定律)/Law of conservation of momentum (動量守恆定律)

A body remains at rest, called at translational equilibrium, or in motion with a constant momentum, unless it is acted upon by a force.

iv Newton's second law of motion (牛頓第二運動定律)

At any instant of time, the net force on a body is equal to the rate at which the body's momentum is changing with time.

v Newton's third law of motion (牛頓第三運動定律)/Law of action force and reaction force (作用力與反作用力定律)

If two bodies exert forces on each other, these forces have the same magnitude but opposite directions.

vi Balanced force

A group of external forces acting on the same system whose sum is zero.

vii Constant acceleration motion

$$\mathbf{x} = \mathbf{x}_0 + \mathbf{v}_0 t + \frac{\mathbf{a}}{2} t^2$$
$$\mathbf{v} = \mathbf{v}_0 + \mathbf{a}t$$

viii Terminal velocity (終端速度)/limiting velocity (極限速度)

The terminal velocity or limiting velocity is the speed of any system subjected to a usually constant force and a velocity-dependent resistance force such that the two forces balance. Let the velocity of the system be $\mathbf{v}$, the constant force be $\mathbf{F}_C$ and parallel to $\mathbf{v}$, and the resistance force $\mathbf{F}_D$ obey:

$$\mathbf{F}_D = -f\left(\mathbf{v} \cdot \hat{\mathbf{F}}_c\right)\hat{\mathbf{v}},$$

where $f : U \subseteq \mathbb{R} \to \mathbb{R}$ is a function.

Then, all v_t such that

$$f(v_t) = \left|\mathbf{F}_C\right|$$

are terminal velocity, and the terminal velocity v_t for one such system with initial velocity $\mathbf{v}_0$ is

$$v_t = g^{-1}\left(\left|\mathbf{F}_C\right|\right),$$

where $g : V \subseteq U \to \mathbb{R}$ is defined as a one-to-one function such that $g = f$ for all $v \in V$ and $\mathbf{v}_0 \cdot \hat{\mathbf{F}}_c \in V$.

IV Work and kinetic energy

i Scalars

- **Work (功)** $W = \int_{\ell} \mathbf{F} \cdot \mathrm{d}\mathbf{r}$. The sum of the works done by multiple forces acting on a system equals the work done by the sum of them. The work done by all forces acting on a system is called net force.
- **Power (功率)** $P = \dfrac{\mathrm{d}W}{\mathrm{d}t} = \mathbf{F} \cdot \mathbf{v}$
- **Kinetic energy (動能)** $K = \dfrac{1}{2}mv^2$

ii Work–energy principle (功能定理)

The work done by all forces acting on a system equals the change in the kinetic energy of the system.

V Potential energy and conservation of mechanical energy

i Conservative force (保守力)

A force that is a conservative field with respect to position.

ii Potential energy (位能)

Potential energy is the energy of an object or system due to the body's position relative to other objects, or the configuration of its particles.

iii Mechanical energy (力學能)

The sum of kinetic and potential energies.

iv Conservation of mechanical energy (力學能守恆)

If no non-conservative force does work on a system, the mechanical energy of the system is a constant.

v Equilibrium point

Given a differentiable conservative force **F**, its potential U, and an equilibrium point $\mathbf{r}_0$. The behavior of a particle at $\mathbf{r}_0$ when an infinisimal displacement occurs is governed by the classification of $\mathbf{r}_0$:

- Stable equilibrium: **F** pushes it back.
- Unstable equilibrium: **F** pushes it away.
- Neutral or indifferent equilibrium: It stay where it becomes.

VI Center of mass (COM or CM) (質心)

i (Mass) density

The density $\rho(\mathbf{r})$ at a point in a system is a scalar field defined as the derivative of mass m with respect to volume V at that point, that is, $\rho(\mathbf{r}) = \dfrac{\mathrm{d}m}{\mathrm{d}V}$.

For a system of N particles with masses m_i at position $\mathbf{r}_i$, the density field $\rho(\mathbf{r})$ is:

$$\rho(\mathbf{r}) = \sum_{i=1}^{N} m_i \delta(\mathbf{r} - \mathbf{r}_i),$$

where δ is the Dirac delta function.

The average density of a system is the total mass of it divided by the volume of it.

ii Center of mass

The (position of) center of mass of a system is the point that moves as if all of the system's mass were concentrated there.

For a system of volume V and density field $\rho(\mathbf{r})$, the (position of) center of mass $\mathbf{r}_c$ is defined as:

$$\mathbf{r}_c = \frac{\iiint_V \rho(\mathbf{r})\mathbf{r}\,\mathrm{d}V}{\iiint_V \rho(\mathbf{r})\,\mathrm{d}V},$$

where the denominator is the total mass of the system.

For a system of N particles with masses m_i at position $\mathbf{r}_i$, the (position of) center of mass $\mathbf{r}_c$ is:

$$\mathbf{r}_c = \frac{\sum_{i=1}^{N} m_i \mathbf{r}_i}{\sum_{i=1}^{N} m_i},$$

where the denominator is the total mass of the system.

iii Vectors

- **Center of mass velocity**

$$\mathbf{v}_c = \frac{\mathrm{d}\mathbf{r}_c}{\mathrm{d}t} = \frac{\iiint_V \rho(\mathbf{r})\mathbf{v}\,\mathrm{d}V}{\iiint_V \rho(\mathbf{r})\,\mathrm{d}V}$$

- **Center of mass acceleration**

$$\mathbf{a}_c = \frac{\mathrm{d}\mathbf{v}_c}{\mathrm{d}t} = \frac{\iiint_V \rho(\mathbf{r})\mathbf{a}\,\mathrm{d}V}{\iiint_V \rho(\mathbf{r})\,\mathrm{d}V}$$

- **Momentum**

$$\mathbf{p} = m\mathbf{v}_c = \iiint_V \rho(\mathbf{r})\mathbf{v}\,\mathrm{d}V$$

iv Scalars

- **Center of mass kinetic energy**:

$$K_c = \frac{m\mathbf{v}_c^{\,2}}{2}$$

  which is only affected by external forces.

- **Total kinetic energy** K: affected by external and internal forces.
- **Internal (kinetic) energy (內（動）能)** $K - K_c$

v Rigid body or rigid object (剛體)

A solid body with zero deformation.

VII 參考系（Frame of reference, reference frame, or frame）

i 參考系

一個抽象的座標系，其原點、方向和比例已在物理空間中基於一組參考點（reference points）而指定。參考點指在數學和物理上定義的幾何點。

ii 慣性參考系（Inertial frame (of reference)）/伽俐略參考系（Galilean (reference) frame）

指在僅考慮實際力時，遵守牛頓運動定律的參考系。

一個慣性參考系相對於另一個慣性參考系必靜止或行等速度運動。

iii 非慣性參考系（Non-inertial frame (of reference)）

指在僅考慮實際力時，不遵守牛頓運動定律的參考系。

iv 假想力（Fictitious force）/慣性力（inertial force）/假力（pseudo-force）

在非慣性參考系中描述系統運動時，欲使得牛頓第二運動定律成立需要作用於系統的非實際力。

VIII 碰撞（Collision）

i 定義

物體接近時，彼此間產生排斥力。

ii 碰撞的性質

若參與碰撞的所有物體為一孤立系統，則該系統遵循：

- 動量、質心動能：碰撞前 = 碰撞中 = 碰撞後
- 總動能、內動能：碰撞中 < 碰撞前 ≤ 後

iii 兩質點縮減質量（reduced mass）/有效慣性質量/約化質量/減縮質量

二質點質量 m_1、m_2，則兩者之系統的縮減質量 μ 定義為：

$$\mu = \frac{1}{\frac{1}{m_1} + \frac{1}{m_2}} = \frac{m_1 m_2}{m_1 + m_2}.$$

iv 兩質點內動能與縮減質量

兩質點系統縮減質量 $\mu = \frac{m_1 m_2}{m_1 + m_2}$、兩質點相對速率 v，內動能 K_{int} 為：

$$K_{int} = \frac{\mu v^2}{2}.$$

v 兩質點內動能與距離

兩質點距離 x、內動能 K_{int}：

- $\frac{\mathrm{d}x}{\mathrm{d}t}$ 與 $\frac{\mathrm{d}K_{int}}{\mathrm{d}t}$ 同號，
- x 最小時 $K_{int} = 0$

vi 兩質點碰撞的恢復係數（coefficient of restitution）

兩質點碰撞，令碰撞前質點 1 相對於質點 2 接近速度 $\mathbf{v}_{12}$，碰撞後質點 2 相對於質點 1 遠離速度 $\mathbf{v}'_{21}$，則恢復係數 e 定義為：

$$e = \frac{\mathbf{v}'_{21}}{\mathbf{v}_{12}}$$

vii　兩質點一維（完全）彈性碰撞（Elastic collision）

- $e = 1$
- 碰撞期間作用力均為保守力。
- 動量、質心動能、力學能、位能 + 內動能：碰撞前 = 碰撞中 = 碰撞後
- 總動能：最接近時 < 碰撞前 = 碰撞後
- 內動能：0= 最接近時 < 碰撞前 = 碰撞後
- 位能：最接近時 > 碰撞前 = 碰撞後
- 一者相對於另一者的相對速度碰撞前、後量值不變、符號變號。

兩質點質量 m_1、m_2，碰撞前，速度 v_1、v_2，質心速度 v_c，內動能 K_{int}，令兩質點縮減質量 μ，兩者彈性碰撞，恢復係數 $e = 1$，無外力，碰撞後，m_1 之速度 v_1'、m_2 之速度 v_2'、內動能 K_{int}'：

$$\mu = \frac{m_1 m_2}{m_1 + m_2}$$

$$v_c = \frac{m_1 v_1 + m_2 v_2}{m_1 + m_2}$$

$$K_{int} = \frac{\mu \left(v_1 - v_2\right)^2}{2}$$

$$v_1' = 2v_c - v_1 = \frac{(m_1 - m_2)v_1 + 2m_2 v_2}{m_1 + m_2}$$

$$v_2' = 2v_c - v_2 = \frac{(m_2 - m_1)v_2 + 2m_1 v_1}{m_2 + m_1}$$

$$K_{int}' = \frac{\mu \left(v_1' - v_2'\right)^2}{2} = \frac{\mu \left(v_1 - v_2\right)^2}{2} = K_{int}$$

若 $m_1 = m_2 = m$：

$$\mu = \frac{m}{2}$$

$$v_c = \frac{v_1 + v_2}{2}$$

$$K_{int} = \frac{m(v_1 - v_2)^2}{4}$$

$$v_1' = 2v_c - v_1 = v_2$$

$$v_2' = 2v_c - v_2 = v_1$$

$$K_{int}' = \frac{m \left(v_1' - v_2'\right)^2}{4} = \frac{m \left(v_1 - v_2\right)^2}{4} = K_{int}$$

若 $m_1 \gg m_2$：

$$\mu = m_2$$
$$v_c = v_1$$
$$K_{int} = \frac{m_2 \left(v_1 - v_2\right)^2}{2}$$
$$v_1' = v_1$$
$$v_2' = 2v_1 - v_2$$
$$K_{int}' = \frac{m_2 \left(v_1' - v_2'\right)^2}{2} = \frac{m_2 \left(v_1 - v_2\right)^2}{2} = K_{int}$$

若 $v_2 = 0$：

$$\mu = \frac{m_1 m_2}{m_1 + m_2}$$
$$v_c = \frac{m_1 v_1}{m_1 + m_2}$$
$$K_{int} = \frac{\mu {v_1}^2}{2}$$
$$v_1' = 2v_c - v_1 = \frac{(m_1 - m_2)v_1}{m_1 + m_2}$$
$$v_2' = 2v_c = \frac{2m_1 v_1}{m_2 + m_1}$$
$$K_{int}' = \frac{\mu \left(v_1' - v_2'\right)^2}{2} = \frac{\mu {v_1}^2}{2} = K_{int}$$

若 $m_1 \gg m_2$ 且 $v_2 = 0$：

$$\mu = m_2$$
$$v_c = v_1$$
$$K_{int} = \frac{m_2 {v_1}^2}{2}$$
$$v_1' = v_1$$
$$v_2' = 2v_1$$
$$K_{int}' = \frac{m_2 \left(v_1' - v_2'\right)^2}{2} = \frac{m_2 {v_1}^2}{2} = K_{int}$$

若 $m_1 \ll m_2$ 且 $v_2 = 0$：

$$\mu = m_1$$
$$v_c = 0$$
$$K_{int} = \frac{m_1 {v_1}^2}{2}$$
$$v_1' = -v_1$$
$$v_2' = 0$$
$$K_{int}' = \frac{m_1 \left(v_1' - v_2'\right)^2}{2} = \frac{m_1 {v_1}^2}{2} = K_{int}$$

viii 動質點一維彈性碰撞靜質點能量變化

兩質點質量 m_1、m_2，碰撞前動能分別為 K_1、0，令 $r=\frac{m_1}{m_2}$，兩者彈性碰撞，碰撞後，m_1 之動能 K_1'、m_2 之動能 K_2'：

$$\frac{K_2'}{K_1}=\frac{4r}{(1+r)^2}=\frac{m_1m_2}{\left(m_1+m_2\right)^2}$$

$$\frac{K_1'}{K_1}=\left(\frac{1-r}{1+r}\right)^2=\left(\frac{m_1-m_2}{m_1+m_2}\right)^2$$

當 $r=1$，能量傳遞比率 $=1$；當 $r=0.5$ 或 $r=2$，能量傳遞比率 $=\frac{8}{9}$。

ix 兩質點一維完全非彈性碰撞（Completely inelastic collision）

- $e=0$
- 碰撞期間作用力有非保守力。
- 動量、質心動能：碰撞前 = 碰撞中 = 碰撞後
- 總動能、力學能、位能 + 內動能：最接近時 = 碰撞後 < 碰撞前
- 內動能：0= 最接近時 = 碰撞後 < 碰撞前
- 位能：最接近時 = 碰撞前 = 碰撞後
- 碰撞後兩者速度相同。

兩質點質量 m_1、m_2，碰撞前，速度 v_1、v_2，質心速度 v_c，內動能 K_{int}，令兩質點縮減質量 μ，兩者完全非彈性碰撞，恢復係數 $e=0$，無外力，碰撞後，m_1 之速度 v_1'、m_2 之速度 v_2'、內動能 K_{int}'：

$$\mu=\frac{m_1m_2}{m_1+m_2}$$

$$v_c=\frac{m_1v_1+m_2v_2}{m_1+m_2}$$

$$K_{int}=\frac{\mu\left(v_1-v_2\right)^2}{2}$$

$$v_1'=v_c=\frac{m_1v_1+m_2v_2}{m_1+m_2}$$

$$v_2'=v_c=\frac{m_2v_2+m_1v_1}{m_2+m_1}$$

$$K_{int}'=\frac{\mu\left(v_1'-v_2'\right)^2}{2}=0$$

若 $m_1 = m_2 = m$：

$$\mu = \frac{m}{2}$$
$$v_c = \frac{v_1 + v_2}{2}$$
$$K_{int} = \frac{m(v_1 - v_2)^2}{4}$$
$$v_1' = v_c = \frac{v_1 + v_2}{2}$$
$$v_2' = v_c = \frac{v_2 + v_1}{2}$$
$$K_{int}' = \frac{m\left(v_1' - v_2'\right)^2}{4} = 0$$

若 $m_1 \gg m_2$：

$$\mu = m_2$$
$$v_c = v_1$$
$$K_{int} = \frac{m_2\left(v_1 - v_2\right)^2}{2}$$
$$v_1' = v_1$$
$$v_2' = v_1$$
$$K_{int}' = \frac{m_2\left(v_1' - v_2'\right)^2}{2} = 0$$

若 $v_2 = 0$：

$$\mu = \frac{m_1 m_2}{m_1 + m_2}$$
$$v_c = \frac{m_1 v_1}{m_1 + m_2}$$
$$K_{int} = \frac{\mu {v_1}^2}{2}$$
$$v_1' = v_c = \frac{m_1 v_1}{m_1 + m_2}$$
$$v_2' = v_c = \frac{m_1 v_1}{m_2 + m_1}$$
$$K_{int}' = \frac{\mu\left(v_1' - v_2'\right)^2}{2} = 0$$

若 $m_1 \gg m_2$ 且 $v_2 = 0$：

$$\mu = m_2$$
$$v_c = v_1$$
$$K_{int} = \frac{m_2 {v_1}^2}{2}$$
$$v_1' = v_1$$
$$v_2' = v_1$$
$$K_{int}' = \frac{m_2\left(v_1' - v_2'\right)^2}{2} = 0$$

若 $m_1 \ll m_2$ 且 $v_2 = 0$：

$$\mu = m_1$$
$$v_c = 0$$
$$K_{int} = \frac{m_1 {v_1}^2}{2}$$
$$v_1' = 0$$
$$v_2' = 0$$
$$K_{int}' = \frac{m_1 \left(v_1' - v_2'\right)^2}{2} = 0$$

x 兩質點一維非彈性碰撞（Inelastic collision）

- $0 \leq e < 1$
- 碰撞期間作用力有非保守力。
- 動量、質心動能：碰撞前 = 碰撞中 = 碰撞後
- 總動能、力學能、位能 + 內動能：最接近時 ≤ 碰撞後 < 碰撞前
- 內動能：0= 最接近時 ≤ 碰撞後 < 碰撞前
- 位能：最接近時 ≥ 碰撞前 = 碰撞後

xi 兩質點一維碰撞通式

兩質點質量 m_1、m_2，碰撞前，速度 v_1、v_2，質心速度 v_c，內動能 K_{int}，令兩質點縮減質量 μ，兩者碰撞，恢復係數 e，無外力，碰撞後，m_1 之速度 v_1'、m_2 之速度 v_2'、內動能 K_{int}'：

$$\mu = \frac{m_1 m_2}{m_1 + m_2}$$
$$v_c = \frac{m_1 v_1 + m_2 v_2}{m_1 + m_2}$$
$$K_{int} = \frac{\mu \left(v_1 - v_2\right)^2}{2}$$
$$v_1' = (1+e)v_c - ev_1 = \frac{(m_1 - em_2)v_1 + (1+e)m_2 v_2}{m_1 + m_2}$$
$$v_2' = (1+e)v_c - ev_2 = \frac{(m_2 - em_1)v_2 + (1+e)m_1 v_1}{m_2 + m_1}$$
$$K_{int}' = \frac{\mu \left(v_1' - v_2'\right)^2}{2} = \frac{\mu \left(e\left(v_1 - v_2\right)\right)^2}{2} = e^2 K_{int}$$

若 $m_1 = m_2 = m$：

$$\mu = \frac{m}{2}$$
$$v_c = \frac{v_1 + v_2}{2}$$
$$K_{int} = \frac{m(v_1 - v_2)^2}{4}$$
$$v_1' = (1+e)v_c - ev_1 = \frac{(1-e)v_1 + (1+e)v_2}{2}$$
$$v_2' = (1+e)v_c - ev_2 = \frac{(1-e)v_2 + (1+e)v_1}{2}$$
$$K_{int}' = \frac{m\left(v_1' - v_2'\right)^2}{4} = \frac{m\left(e\left(v_1 - v_2\right)\right)^2}{4} = e^2 K_{int}$$

若 $m_1 \gg m_2$：

$$\mu = m_2$$
$$v_c = v_1$$
$$K_{int} = \frac{m_2\left(v_1 - v_2\right)^2}{2}$$
$$v_1' = v_1$$
$$v_2' = (1+e)v_1 - ev_2$$
$$K_{int}' = \frac{m_2\left(v_1' - v_2'\right)^2}{2} = \frac{m_2\left(e\left(v_1 - v_2\right)\right)^2}{2} = e^2 K_{int}$$

若 $v_2 = 0$：

$$\mu = \frac{m_1 m_2}{m_1 + m_2}$$
$$v_c = \frac{m_1 v_1}{m_1 + m_2}$$
$$K_{int} = \frac{\mu {v_1}^2}{2}$$
$$v_1' = (1+e)v_c - ev_1 = \frac{(m_1 - em_2)v_1}{m_1 + m_2}$$
$$v_2' = (1+e)v_c = \frac{(1+e)m_1 v_1}{m_2 + m_1}$$
$$K_{int}' = \frac{\mu\left(v_1' - v_2'\right)^2}{2} = \frac{\mu\left(ev_1\right)^2}{2} = e^2 K_{int}$$

若 $m_1 \gg m_2$ 且 $v_2 = 0$：

$$\mu = m_2$$
$$v_c = v_1$$
$$K_{int} = \frac{m_2 {v_1}^2}{2}$$
$$v_1' = v_1$$
$$v_2' = (1+e)v_1$$
$$K_{int}' = \frac{m_2 \left(v_1' - v_2'\right)^2}{2} = \frac{m_2 \left(ev_1\right)^2}{2} = e^2 K_{int}$$

若 $m_1 \ll m_2$ 且 $v_2 = 0$：

$$\mu = m_1$$
$$v_c = 0$$
$$K_{int} = \frac{m_1 {v_1}^2}{2}$$
$$v_1' = -ev_1$$
$$v_2' = 0$$
$$K_{int}' = \frac{m_1 \left(v_1' - v_2'\right)^2}{2} = \frac{m_1 \left(ev_1\right)^2}{2} = e^2 K_{int}$$

xii　兩質點夾彈簧一維彈性碰撞

兩質點質量 m_1、m_2，碰撞前，速度 v_1、v_2，內動能 K_{int}，令兩質點縮減質量 $\mu = \frac{m_1 m_2}{m_1 + m_2}$，$m_1$ 靠 m_2 側或 m_2 靠 m_1 側黏有一質量不計、力常數為 k 之理想彈簧，兩者彈性碰撞，兩者最近時中彈簧被壓縮之最大長度 x：

$$\frac{kx^2}{2} = \frac{\mu(v_1 - v_2)^2}{2} = K_{int}$$

xiii　兩質點碰撞通式

兩質點質量 m_1、m_2，碰撞前，速度 $\mathbf{v}_1$、$\mathbf{v}_2$，質心速度 $\mathbf{v}_c$，內動能 K_{int}，令兩質點縮減質量 μ：

$$\mu = \frac{m_1 m_2}{m_1 + m_2},$$

令原點在直線 $\mathbf{v}_1 + t(\mathbf{v}_2 - \mathbf{v}_1),\quad t \in \mathbb{R}$ 上的垂足 $\mathbf{v}_t$：

$$\mathbf{v}_t = \frac{\left(\left|\mathbf{v}_2\right|^2 - \mathbf{v}_1 \cdot \mathbf{v}_2\right)\mathbf{v}_1 + \left(\left|\mathbf{v}_1\right|^2 - \mathbf{v}_1 \cdot \mathbf{v}_2\right)\mathbf{v}_2}{\left|\mathbf{v}_1 - \mathbf{v}_2\right|^2},$$

令 m_1 相對 m_2 速度方向單位向量 $\hat{\mathbf{v}}_n$：

$$\hat{\mathbf{v}}_n = \frac{\mathbf{v}_1 - \mathbf{v}_2}{\left|\mathbf{v}_1 - \mathbf{v}_2\right|},$$

令 $\mathbf{v}_1$ 與 $\mathbf{v}_2$ 在 $\hat{\mathbf{v}}_n$ 方向分量 v_1, v_2：

$$v_1 = \mathbf{v}_1 \cdot \hat{\mathbf{v}}_n,$$

$$v_2 = \mathbf{v}_2 \cdot \hat{\mathbf{v}}_n,$$

兩者碰撞，恢復係數 e，無外力，碰撞後，m_1 之速度 $\mathbf{v}_1'$、m_2 之速度 $\mathbf{v}_2'$，令 $\mathbf{v}_1'$ 與 $\mathbf{v}_2'$ 在 $\hat{\mathbf{v}}_n$ 方向分量 v_1', v_2'：

$$v_1' = \mathbf{v}_1' \cdot \hat{\mathbf{v}}_n,$$

$$v_2' = \mathbf{v}_2' \cdot \hat{\mathbf{v}}_n,$$

內動能 K_{int}'：

$$\mathbf{v}_1 = \mathbf{v}_t + v_1 \hat{\mathbf{v}}_n$$

$$\mathbf{v}_2 = \mathbf{v}_t + v_2 \hat{\mathbf{v}}_n$$

$$\mathbf{v}_c = \mathbf{v}_t + \frac{m_1 v_1 + m_2 v_2}{m_1 + m_2} \hat{\mathbf{v}}_n$$

$$K_{int} = \frac{\mu \left(v_1 - v_2\right)^2}{2}$$

$$v_1' = (1+e)v_c - e v_1 = \frac{(m_1 - e m_2)v_1 + (1+e)m_2 v_2}{m_1 + m_2}$$

$$v_2' = (1+e)v_c - e v_2 = \frac{(m_2 - e m_1)v_2 + (1+e)m_1 v_1}{m_2 + m_1}$$

$$\mathbf{v}_1' = \mathbf{v}_t + v_1' \hat{\mathbf{v}}_n$$

$$\mathbf{v}_2' = \mathbf{v}_t + v_2' \hat{\mathbf{v}}_n$$

$$K_{int}' = \frac{\mu \left(v_1' - v_2'\right)^2}{2} = \frac{\mu \left(e \left(v_1 - v_2\right)\right)^2}{2} = e^2 K_{int}$$

IX　變質量系統與火箭方程（Rocket Equations）

i　質量增加

令無外力下：

- 時間 t 時：主系統質量 m、速度 $\mathbf{v}$，副系統質量 $\mathrm{d}m$、速度 $\mathbf{v} + \mathbf{v}_{rel}$，兩系統總動量 $\mathbf{p}$；
- 時間 $(t + \mathrm{d}t)$ 時：副系統併入主系統，主系統的性質在 t 時為 i 者變為 $i + \mathrm{d}i$，主系統動量仍為 $\mathbf{p}$。

$$\begin{aligned}\mathbf{p} &= m\mathbf{v} + (\mathrm{d}m)\left(\mathbf{v} + \mathbf{v}_{rel}\right) \\ &= \left(m + \mathrm{d}m\right)(\mathbf{v} + \mathrm{d}\mathbf{v})\end{aligned}$$

故：

$$m(\mathrm{d}\mathbf{v}) - (\mathrm{d}m)\mathbf{v}_{rel} = 0.$$

令質量增加率 $R = \dfrac{\mathrm{d}m}{\mathrm{d}t}$，加速度 $\mathbf{a} = \dfrac{\mathrm{d}\mathbf{v}}{\mathrm{d}t}$：

$$R\mathbf{v}_{rel} = m\mathbf{a}.$$

令時間 t_1 與 t_2 時 $\mathbf{v}$ 與 m 分別為 $\mathbf{v}_1$、m_1 與 $\mathbf{v}_2$、m_2。積分前式：

$$\mathbf{v}_2 - \mathbf{v}_1 = \mathbf{v}_{rel} \int_{t_1}^{t_2} \frac{R}{m} \mathrm{d}t \,.$$

由於：

$$\frac{\mathrm{d} \ln |f(x)|}{\mathrm{d}x} = \frac{f'(x)}{f(x)}.$$

我們有：

$$\mathbf{v}_2 - \mathbf{v}_1 = \mathbf{v}_{rel} \ln \frac{M_2}{M_1}.$$

ii 質量減少與火箭方程

令無外力下：

- 時間 t 時：主系統質量 m、速度 $\mathbf{v}$、動量 $\mathbf{p}$；
- 時間 $(t + \mathrm{d}t)$ 時：主系統分出質量 $\mathrm{d}m$ 的副系統，使得主系統質量變為 $m - \mathrm{d}m$，主系統的質量以外性質在 t 時為 i 者變為 $i + \mathrm{d}i$，副系統速度 $\mathbf{v} + \mathbf{v}_{rel}$，主系統動量仍為 $\mathbf{p}$。

$$\begin{aligned} \mathbf{p} &= m\mathbf{v} \\ &= \left(m - \mathrm{d}m\right)(\mathbf{v} + \mathrm{d}\mathbf{v}) + \mathrm{d}m\left(\mathbf{v} + \mathbf{v}_{rel}\right) \end{aligned}$$

故：

$$\left(m - \mathrm{d}m\right)(\mathrm{d}\mathbf{v}) + (\mathrm{d}m)\mathbf{v}_{rel} = 0.$$

考慮 $(\mathrm{d}m)(\mathrm{d}\mathbf{v}) = 0$，故：

$$m(\mathrm{d}\mathbf{v}) + (\mathrm{d}m)\mathbf{v}_{rel} = 0.$$

令質量減少率 $R = -\dfrac{\mathrm{d}m}{\mathrm{d}t}$，加速度 $\mathbf{a} = \dfrac{\mathrm{d}\mathbf{v}}{\mathrm{d}t}$：

$$R\mathbf{v}_{rel} = m\mathbf{a}.$$

對於火箭，我們稱 R 為燃料消耗質量率（mass rate of fuel consumption），稱 $R\mathbf{v}_{rel}$ 為推力（thrust），稱此方程為火箭第一方程（First rocket equation）。

令時間 t_1 與 t_2 時 $\mathbf{v}$ 與 m 分別為 $\mathbf{v}_1$、m_1 與 $\mathbf{v}_2$、m_2。積分前式：

$$\mathbf{v}_2 - \mathbf{v}_1 = \mathbf{v}_{rel} \int_{t_1}^{t_2} \frac{R}{m} \mathrm{d}t \,.$$

由於：

$$\frac{\mathrm{d} \ln |f(x)|}{\mathrm{d}x} = \frac{f'(x)}{f(x)}.$$

我們有：

$$\mathbf{v}_2 - \mathbf{v}_1 = -\mathbf{v}_{rel} \ln \frac{M_2}{M_1}.$$

稱此方程為火箭第二方程（Second rocket equation）。

X Rotation (轉動)

i Rotation

Rotations is a rigid body movement which, unlike a translation, keeps at least one point fixed. We call a rotation to be around/about the point, or axis when all points on a line are fixed, and call such point/axis point/axis of the rotation or rotational/fixed point/axis.

All rigid body movements are rotations, translations, or combinations of the two.

ii Euler's rotation theorem and rotational matrix

Any motion of a rigid body such that a point on it remains fixed to the body, is equivalent to a single rotation about some axis that passes through the fixed point.

The sum of finite rotations around a same point or axis is also a rotation. The inverse of a rotation is also a rotation. Thus, the rotations around a point or axis form a group, which is, when expressed in rotational matrices with a fixed point chosen to be the origin of the three-dimensional Euclidean vector space, the special orthogonal group of degree three $\mathrm{SO}(3)$ over $\mathbb{R}$.

However, a rotation around a point or axis and a rotation around a different point or axis may result in something other than a rotation, that is, a combination of a rotation and a nonzero translation.

iii Principal rotations

Rotations around the x, y and z axes are called principal rotations. Rotation around any axis can be performed by taking a rotation around the x axis, followed by a rotation around the y axis, and followed by a rotation around the z axis. That is, any spatial rotation can be decomposed into a combination of principal rotations.

iv (Instantaneous) angular velocity or angular frequency (（瞬時）角速度/角頻率)

If a rigid body is rotating about a fixed point and a point in the rigid body at position $\mathbf{r}$ from the fixed point is at velocity $\dot{\mathbf{r}}$, then the (instantaneous) angular velocity or angular frequency $\boldsymbol{\omega}$ of the body is defined with

$$\dot{\mathbf{r}} = \boldsymbol{\omega} \times \mathbf{r}.$$

The (instantaneous) angular speed or angular frequency $\omega = |\boldsymbol{\omega}|$.

The (instantaneous) angular acceleration $\boldsymbol{\alpha} = \dfrac{\mathrm{d}\boldsymbol{\omega}}{\mathrm{d}t}$.

v Moment of inertia or rotational inertia

The moment of inertia or rotational inertia $\mathbf{I}$ of an object of volume V is a measure of the object's resistance to changes to its rotation, defined with the density field $\rho(\mathbf{r})$:

$$\mathbf{I} = \begin{pmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{yx} & I_{yy} & I_{yz} \\ I_{zx} & I_{zy} & I_{zz} \end{pmatrix},$$

where:

$$\begin{aligned}
I_{xx} &= \iiint_V \rho(\mathbf{r}) \left(\left(\mathbf{r} \cdot \hat{\mathbf{j}} \right)^2 + \left(\mathbf{r} \cdot \hat{\mathbf{k}} \right)^2 \right) \mathrm{d}V, \\
I_{yy} &= \iiint_V \rho(\mathbf{r}) \left(\left(\mathbf{r} \cdot \hat{\mathbf{i}} \right)^2 + \left(\mathbf{r} \cdot \hat{\mathbf{k}} \right)^2 \right) \mathrm{d}V, \\
I_{zz} &= \iiint_V \rho(\mathbf{r}) \left(\left(\mathbf{r} \cdot \hat{\mathbf{i}} \right)^2 + \left(\mathbf{r} \cdot \hat{\mathbf{j}} \right)^2 \right) \mathrm{d}V, \\
I_{xy} = I_{yx} &= - \iiint_V \rho(\mathbf{r}) \left(\mathbf{r} \cdot \hat{\mathbf{i}} \right) \cdot \left(\mathbf{r} \cdot \hat{\mathbf{j}} \right) \mathrm{d}V, \\
I_{yz} = I_{zy} &= - \iiint_V \rho(\mathbf{r}) \left(\mathbf{r} \cdot \hat{\mathbf{j}} \right) \cdot \left(\mathbf{r} \cdot \hat{\mathbf{z}} \right) \mathrm{d}V, \\
I_{xz} = I_{zx} &= - \iiint_V \rho(\mathbf{r}) \left(\mathbf{r} \cdot \hat{\mathbf{i}} \right) \cdot \left(\mathbf{r} \cdot \hat{\mathbf{z}} \right) \mathrm{d}V,
\end{aligned}$$

where $\hat{\mathbf{i}}, \hat{\mathbf{j}}, \hat{\mathbf{k}}$ are the unit vectors in the positive x-, y-, and z-axes.

We can find a set of x-, y-, and z-axes such that $I_{xy} = I_{xz} = I_{yz} = 0$, that is, $\mathbf{I}$ is diagonal. In this case, we define

$$I_1 = I_{xx}, \quad I_2 = I_{yy}, \quad I_3 = I_{zz}.$$

vi Angular momentum (角動量)

The angular momentum $\mathbf{L}$ is

$$\mathbf{L} = \mathbf{I}\boldsymbol{\omega} = \iiint_V \rho \mathbf{r} \mathbf{r} \times \dot{\mathbf{r}} \, \mathrm{d}V.$$

vii Torque (力矩)

The torque $\boldsymbol{\tau}$ is the twisting effect of a force F applied to a rigid body at position $\mathbf{r}$ from a fixed point of rotation given by

$$\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F},$$

where $\mathbf{r}$ is called the lever arm (力臂) of the force, and $\mathbf{r} \times \hat{\mathbf{F}} \times \hat{\mathbf{F}}$ or its magnitude, often denoted $r_\perp$, is called the effective lever arm of the force.

Torque and angular momentum are related according to

$$\boldsymbol{\tau} = \frac{\mathrm{d}\mathbf{L}}{\mathrm{d}t},$$

called Newton's second law for rotation. Thus, if $\mathbf{I}$ is a constant, torque and angular acceleration are related according to

$$\boldsymbol{\tau} = \mathbf{I}\boldsymbol{\alpha}.$$

viii Power, work, and kinetic energy

The power pf P of a torque $\boldsymbol{\tau}$ is

$$P = \boldsymbol{\tau} \cdot \boldsymbol{\omega}$$

The work W done by a torque $\boldsymbol{\tau}$ in some time interval t_0 to t_1 is

$$W = \int_{t_0}^{t_1} \boldsymbol{\tau} \cdot \boldsymbol{\omega} \, \mathrm{d}t \, .$$

The kinetic energy K_r due to the rotation, called rotational kinetic energy, is given by

$$K_r = \frac{1}{2}\boldsymbol{\omega}^{\top}\mathbf{I}\boldsymbol{\omega}.$$

The translational kinetic energy K_t associated with the motion of the rotation axis is given by

$$K_t = \frac{1}{2}m{v_r}^2$$

where m is the total mass of the rigid body and v_r is the speed of the axis of rotation.

The total kinetic energy K of the rigid body is given by

$$K = \frac{1}{2}\iint_V \rho\mathbf{r}\,|\dot{\mathbf{r}}|^2 \, \mathrm{d}V = K_r + K_t.$$

ix Angular position

For rotations around only one axis, we defined the angular position of a point **P** about a given reference axis, which is a line oriented along a unit vector $\hat{\mathbf{n}}$, relative to a given polar axis on some plane F perpendicular to the reference axis, which defines a polar coordinate system whose positive sense is determined by looking upward along $\hat{\mathbf{n}}$ (right-hand rule), as the polar angle θ (one-dimensional) of the projection of **P** on F or $\boldsymbol{\theta} = \theta\hat{\mathbf{n}}$ (three-dimensional).

The angular displacement $\Delta\theta$ (one-dimensional) or $\Delta\boldsymbol{\theta}$ (three-dimensional) is the angular position at the end subtracted by the angular position at the beginning, where the reference axis is the fixed axis in the rotation. Angular displacements are additive for rotations about the same axis. However, they are not additive for rotations about different axes.

The three-dimensional quantities defined here are the same as the ones defined earlier if only rotation about that axis exists, that is, they are parallel to $\hat{\mathbf{n}}$.

The (instantaneous) angular velocity ω (one-dimensional) and $\boldsymbol{\omega}$ (three-dimensional) are defined as

$$\omega = \frac{\mathrm{d}\theta}{\mathrm{d}t}$$

$$\boldsymbol{\omega} = \frac{\mathrm{d}\boldsymbol{\theta}}{\mathrm{d}t}$$

The (instantaneous) angular acceleration α (one-dimensional) and $\boldsymbol{\alpha}$ (three-dimensional) are defined as

$$\alpha = \frac{\mathrm{d}\omega}{\mathrm{d}t}$$

$$\boldsymbol{\alpha} = \frac{\mathrm{d}\boldsymbol{\omega}}{\mathrm{d}t}$$

The moment of inertia or rotational inertia as a scalar I of an object of volume V about its rotational axis is defined, with the density field $\rho(\mathbf{r})$ over the position space where some point on the axis is origin and the unit vector in the direction of axis is $\hat{\mathbf{n}}$, as

$$I = \iiint_V \rho(\mathbf{r}) \left|\mathbf{r} \times \hat{\mathbf{n}}\right|^2 \mathrm{d}V \,.$$

The angular momentum L (one-dimensional) and $\mathbf{L}$ (three-dimensional) are defined as

$$L = I\omega$$

$$\mathbf{L} = I\boldsymbol{\omega}$$

Let $\mathbf{r}_r$ be the orthogonal projection of $\mathbf{r}$ on F:

$$\mathbf{r}_r = \mathbf{r} - \mathbf{r} \cdot \hat{\mathbf{n}}\hat{\mathbf{n}}$$

The torque τ (one-dimensional) and $\boldsymbol{\tau}$ (three-dimensional) for force $\mathbf{F}$ perpendicular to $\hat{\mathbf{n}}$ (otherwise the rotation is not purely on this axis) are defined as

$$\boldsymbol{\tau} = \mathbf{r}_r \times \mathbf{F}$$

$$\tau = \boldsymbol{\tau} \cdot \hat{\mathbf{n}}$$

and satisfy:

$$\boldsymbol{\tau} = \frac{\mathrm{d}\mathbf{L}}{\mathrm{d}t}$$

$$\tau = \frac{\mathrm{d}L}{\mathrm{d}t}$$

$$\boldsymbol{\tau} = \mathbf{I}\boldsymbol{\alpha} = I\boldsymbol{\alpha}$$

$$\tau = I\alpha$$

$$P = \boldsymbol{\tau} \cdot \boldsymbol{\omega}$$

$$P = \tau\omega$$

The work W done by a torque τ is

$$W = \int_{\theta_1}^{\theta_2} \tau \, \mathrm{d}\theta$$

The rotational kinetic energy is

$$K_r = \frac{1}{2} I\omega^2$$

x Planar lamina or plane lamina (薄片)

A planar lamina or plane lamina is a planar object that is infinitely thin.

xi Perpendicular axis theorem (垂直軸定理或正交軸定理) or plane figure theorem

Define axes x, y, z perpendicular to each other and meet at origin O. Suppose a planar lamina rigid body lies on the xy plane. Let I_x, I_y, I_z be moments of inertia about axis x, y, z respectively, then the perpendicular axis theorem states that

$$I_z = I_x + I_y.$$

Proof.

$$I_z = \int (x^2 + y^2)\,\mathrm{d}m = \int x^2\,\mathrm{d}m + \int y^2\,\mathrm{d}m = I_y + I_x$$

□

xii Parallel axis theorem (平行軸定理), Huygens–Steiner theorem, or Steiner's theorem

Suppose a rigid body of mass M is rotated about an axis z passing through the body's center of mass. The body has a moment of inertia I_{cm} with respect to this axis. The parallel axis theorem states that if the body is made to rotate instead about a new axis z', which is parallel to the first axis and displaced from it by a distance D, then the moment of inertia I with respect to the new axis is related to I_{cm} by

$$I = I_{cm} + MD^2.$$

Proof.

$$\begin{aligned} I &= \int \left((x - D)^2 + y^2\right)\mathrm{d}m \\ &= \int (x^2 + y^2)\,\mathrm{d}m + D^2 \int \mathrm{d}m - 2D \int x\,\mathrm{d}m \\ &= I_{cm} + MD^2 \end{aligned}$$

□

xiii Law of conservation of angular momentum (角動量守恆定律)

A body remains without rotation, called at rotational equilibrium, or in rotation with a constant angular momentum, unless it is acted upon by a torque.

xiv Balanced torque

A group of torque forces acting on the same rigid body whose sum is zero.

xv Rotational equilibrium under three forces

一剛體受三力達轉動平衡，則三力延長線必交於一點。

xvi Static equilibrium

A system in both translational and rotational equilibrium is called in static equilibrium.

xvii Fulcrum (支點)

A point in a rigid body on which external force is applied such that the position of it is fixed.

xviii Spinning top or top

A spinning top or simply top is a rigid body of which the mass is distributed axially symmetric about the rotational axis fixed to the body.

xix Moment of inertia of a hoop about its central axis

A uniform rigid infinitely thin hoop or infinitely thin annular cylinder of mass M and radius R has moment of inertia I about its central axis:

$$I = MR^2.$$

xx Moment of inertia of a hoop about any axis passing through its center of mass and perpendicular to its central axis

A uniform rigid infinitely thin hoop of mass M and radius R has moment of inertia I about any axis passing through its center of mass and perpendicular to its central axis:

$$I = \frac{MR^2}{2}.$$

Proof. By perpendicular axis theorem and the symmetry of the object, $2I = MR^2$. □

xxi Moment of inertia of a disk about its central axis

A uniform rigid infinitely thin disk of mass M and radius R has moment of inertia I about its central axis:

$$I = \frac{MR^2}{2}.$$

Proof.

$$\begin{aligned} I &= \frac{M}{\pi R^2}\int_0^R r^2 2\pi r \,\mathrm{d}r \\ &= \frac{2M}{R^2}\int_0^R r^3 \,\mathrm{d}r \\ &= \frac{MR^2}{2} \end{aligned}$$

□

xxii Moment of inertia of a disk about any axis passing through its center of mass and perpendicular to its central axis

A uniform rigid infinitely thin disk of mass M and radius R has moment of inertia I about any axis passing through its center of mass and perpendicular to its central axis:

$$I = \frac{MR^2}{4}.$$

Proof. By perpendicular axis theorem and the symmetry of the object, $2I = \frac{MR^2}{2}$. □

xxiii Moment of inertia of a cylinder about its central axis

A uniform rigid cylinder of mass M and radius R has moment of inertia I about its central axis:

$$I = \frac{MR^2}{2}.$$

Proof.

$$\begin{aligned} I &= \frac{M}{\pi R^2} \int_0^R r^2 2\pi r \, \mathrm{d}r \\ &= \frac{2M}{R^2} \int_0^R r^3 \, \mathrm{d}r \\ &= \frac{MR^2}{2} \end{aligned}$$

□

xxiv Moment of inertia of a cylinder about any axis passing through its center of mass and perpendicular to its central axis

A uniform rigid cylinder of mass M, radius R, and length L has moment of inertia I about any axis passing through its center of mass and perpendicular to its central axis:

$$I = \frac{M}{12}\left(3R^2 + L^2\right).$$

Proof.

$$
\begin{aligned}
I &= \frac{2M}{\pi R^2 L} \int_0^{\frac{L}{2}} \int_0^R \int_0^{2\pi} (r^2 \sin^2\theta + l^2) r \,\mathrm{d}\theta \,\mathrm{d}r \,\mathrm{d}l \\
&= \frac{2M}{\pi R^2 L} \int_0^{\frac{L}{2}} \int_0^R \left(r^3 \left(\frac{\theta}{2} - \frac{\sin(2\theta)}{4} \right) + l^2 r\theta \right) \Big|_0^{2\pi} \mathrm{d}r \,\mathrm{d}l \\
&= \frac{2M}{\pi R^2 L} \int_0^{\frac{L}{2}} \int_0^R \left(\pi r^3 + 2\pi l^2 r \right) \mathrm{d}r \,\mathrm{d}l \\
&= \frac{2M}{R^2 L} \int_0^{\frac{L}{2}} \frac{R^4}{4} + l^2 R^2 \,\mathrm{d}l \\
&= \frac{2M}{L} \int_0^{\frac{L}{2}} \frac{R^2}{4} + l^2 \,\mathrm{d}l \\
&= \frac{2M}{L} \left(\frac{R^2 L}{8} + \frac{L^3}{24} \right) \\
&= \frac{M}{12} \left(3R^2 + L^2 \right)
\end{aligned}
$$

□

xxv Moment of inertia of a thin rod about any axis passing through its center of mass and perpendicular to its central axis

A uniform thin rod of mass M and length L has moment of inertia I about any axis passing through its center of mass and perpendicular to its central axis:

$$I = \frac{ML^2}{12}.$$

Proof.

$$
\begin{aligned}
I &= 2 \int_0^{\frac{L}{2}} l^2 \frac{M}{L} \,\mathrm{d}l \\
&= \frac{2M}{L} \frac{L^3}{24} \\
&= \frac{ML^2}{12}
\end{aligned}
$$

□

xxvi Moment of inertia of a spherical shell about any axis passing through its center of mass

A uniform rigid spherical shell of mass M and radius R has moment of inertia I about any axis passing through its center of mass:

$$I = \frac{2MR^2}{3}.$$

Proof.

$$\begin{aligned} I &= \frac{M}{4\pi R^2} \int_0^{\pi} \int_0^{2\pi} R^2 \sin^2\theta R^2 \sin\theta \, \mathrm{d}\phi \, \mathrm{d}\theta \\ &= \frac{MR^2}{4\pi} \int_0^{\pi} \int_0^{2\pi} \sin^3\theta \, \mathrm{d}\phi \, \mathrm{d}\theta \\ &= \frac{MR^2}{2} \int_0^{\pi} \sin^3\theta \, \mathrm{d}\theta \\ &= \frac{MR^2}{3} \left(-\cos\theta\right) \Big|_0^{\pi} \\ &= \frac{2MR^2}{3} \end{aligned}$$

□

xxvii Moment of inertia of a sphere about any axis passing through its center of mass

A uniform rigid sphere of mass M and radius R has moment of inertia I about any axis passing through its center of mass:

$$I = \frac{2MR^2}{5}.$$

Proof.

$$\begin{aligned} I &= \frac{M}{\frac{4}{3}\pi R^3} \int_0^{R} \int_0^{\pi} \int_0^{2\pi} r^2 \sin^2\theta r^2 \sin\theta \, \mathrm{d}\phi \, \mathrm{d}\theta \, \mathrm{d}r \\ &= \frac{3M}{4\pi R^3} \int_0^{R} \int_0^{\pi} \int_0^{2\pi} r^4 \sin^3\theta \, \mathrm{d}\phi \, \mathrm{d}\theta \, \mathrm{d}r \\ &= \frac{3M}{2R^3} \int_0^{R} \int_0^{\pi} r^4 \sin^3\theta \, \mathrm{d}\theta \, \mathrm{d}r \\ &= \frac{3M}{2R^3} \int_0^{R} r^4 \left(-\frac{1}{3}\sin^2\theta\cos\theta - \frac{2}{3}\cos\theta\right) \Big|_0^{\pi} \mathrm{d}r \\ &= \frac{2M}{R^3} \int_0^{R} r^4 \, \mathrm{d}r \\ &= \frac{2MR^2}{5} \end{aligned}$$

□

xxviii Moment of inertia of a cuboid about axis passing through its center of mass and perpendicular to its surface

A $a \times b \times c$ uniform rigid cuboid of mass M has moment of inertia I about the axis passing through its center of mass and perpendicular to the $a \times b$ surfaces of it:

$$I = \frac{M}{12}\left(a^2 + b^2\right).$$

Proof.

$$\begin{aligned} I &= \frac{4M}{ab} \int_0^{\frac{a}{2}} \int_0^{\frac{b}{2}} A^2 + B^2 \,\mathrm{d}B \,\mathrm{d}A \\ &= \frac{4M}{ab} \int_0^{\frac{a}{2}} \frac{A^2 b}{2} + \frac{b^3}{24} \,\mathrm{d}A \\ &= \frac{M}{6a} \int_0^{\frac{a}{2}} 12A^2 + b^2 \,\mathrm{d}A \\ &= \frac{M}{6a} \left(\frac{a^3}{2} + \frac{ab^2}{2} \right) \\ &= \frac{M}{12} \left(a^2 + b^2 \right) \end{aligned}$$

□

xxix Moment of inertia of a cone about its central axis

A uniform rigid cone of mass M, radius of bottom R, and length L has moment of inertia I about its central axis:

$$I = \frac{3MR^2}{10}.$$

Proof.

$$\begin{aligned} I &= \frac{M}{\frac{\pi R^2 L}{3}} \int_0^L \int_0^{\frac{Rl}{L}} r^2 2\pi r \,\mathrm{d}r \,\mathrm{d}l \\ &= \frac{6M}{R^2 L} \int_0^L \int_0^{\frac{Rl}{L}} r^3 \,\mathrm{d}r \,\mathrm{d}l \\ &= \frac{6M}{R^2 L} \int_0^L \frac{1}{4} \left(\frac{Rl}{L} \right)^4 \mathrm{d}l \\ &= \frac{3MR^2}{2L^5} \int_0^L l^4 \,\mathrm{d}l \\ &= \frac{3MR^2}{2L^5} \frac{L^5}{5} \\ &= \frac{3MR^2}{10} \end{aligned}$$

□

xxx Moment of inertia of a cone about any axis passing through its center of mass and perpendicular to its central axis

A uniform rigid cone of mass M, radius of bottom R, and length L has moment of inertia I about any axis passing through its center of mass and perpendicular to its central axis:

$$I = \frac{3M}{80} \left(4R^2 + L^2 \right).$$

Proof.

$$\frac{\int_0^L l\pi R^2 \left(\frac{l}{L}\right)^2 \mathrm{d}l}{\int_0^L \pi R^2 \left(\frac{l}{L}\right)^2 \mathrm{d}l} = \frac{\int_0^L l^3 \,\mathrm{d}l}{\int_0^L l^2 \,\mathrm{d}l} = \frac{3L}{4}$$

Thus, the center of mass of it is on the central axis of it and $\frac{3L}{4}$ below its vertex.

$$\begin{aligned} I &= \frac{3M}{\pi R^2 L} \int_0^L \left(\frac{R^2}{4}\frac{l^2}{L^2} + \left(l - \frac{3L}{4}\right)^2\right) \pi R^2 \frac{l^2}{L^2} \,\mathrm{d}l \\ &= \frac{3M}{16L^5} \int_0^L 4R^2 l^4 + 16L^2 l^4 - 24L^3 l^3 + 9L^4 l^2 \,\mathrm{d}l \\ &= \frac{3M}{80}\left(4R^2 + L^2\right) \end{aligned}$$

□

xxxi Moment of inertia of a hollow cone about its central axis

A uniform rigid hollow cone of mass M, radius of bottom R, and length L has moment of inertia I about its central axis:

$$I = \frac{MR^2}{2}.$$

Proof.

$$\begin{aligned} I &= \frac{M}{\int_0^L 2\pi R \frac{l}{L} \,\mathrm{d}l} \int_0^L R^2 \frac{l^2}{L^2} 2\pi R \frac{l}{L} \,\mathrm{d}l \\ &= \frac{2MR^2}{L^4} \int_0^L l^3 \,\mathrm{d}l \\ &= \frac{MR^2}{2} \end{aligned}$$

□

xxxii Moment of inertia of a hollow cone about any axis passing through its center of mass and perpendicular to its central axis

A uniform rigid hollow cone of mass M, radius of bottom R, and length L has moment of inertia I about any axis passing through its center of mass and perpendicular to its central axis:

$$I = \frac{M}{36}\left(9R^2 + 2L^2\right).$$

Proof.

$$\frac{\int_0^L l 2\pi R \frac{l}{L} \,\mathrm{d}l}{\int_0^L 2\pi R \frac{l}{L} \,\mathrm{d}l} = \frac{\frac{L^3}{3}}{\frac{L^2}{2}} = \frac{2L}{3}.$$

Thus, the center of mass of it is on the central axis of it and $\frac{2L}{3}$ below its vertex.

$$\begin{aligned}
I &= \frac{M}{\int_0^L 2\pi R\frac{l}{L}\,\mathrm{d}l}\int_0^L \left(\frac{1}{2}R^2\left(\frac{l}{L}\right)^2+\left(l-\frac{2L}{3}\right)^2\right)2\pi R\frac{l}{L}\,\mathrm{d}l \\
&= \frac{M}{\pi RL}\int_0^L \left(\frac{1}{2}R^2\left(\frac{l}{L}\right)^2+l^2-\frac{4Ll}{3}+\frac{4L^2}{9}\right)2\pi R\frac{l}{L}\,\mathrm{d}l \\
&= \frac{M}{L^4}\int_0^L \left(R^2l^2+2L^2l^2-\frac{8L^3l}{3}+\frac{8L^4}{9}\right)l\,\mathrm{d}l \\
&= \frac{M}{9L^4}\int_0^L 9R^2l^3+18L^2l^3-24L^3l^2+8L^4l\,\mathrm{d}l \\
&= \frac{M}{36}\left(9R^2+18L^2-32L^2+16L^2\right) \\
&= \frac{M}{36}\left(9R^2+2L^2\right)
\end{aligned}$$

□

xxxiii 等速率圓周運動（Uniform circular motion）

等速率圓周運動指一個二維週期性運動，其中位置 **x** 符合：

$$\ddot{\mathbf{x}} = \boldsymbol{\omega}\times(\boldsymbol{\omega}\times(\mathbf{x}-\mathbf{C}))$$

其中 $\boldsymbol{\omega}$ 和 **C** 是常向量。

軌跡為一以 **C** 為圓心的圓，稱軌跡圓，運動過程中速率與加速度量值均始終不變，並做繞通過 **C** 垂直軌跡圓之軸之轉動，角速度 $\boldsymbol{\omega}$，角速率 $\omega = |\boldsymbol{\omega}|$，週期 $\frac{2\pi}{\omega}$，加速度、速度與角速度始終兩兩相垂直。質量 m 質點做等速率圓周運動時，稱 $m\ddot{\mathbf{x}}$ 為向心力（centripetal force）。

若以軌跡圓圓心為原點在軌跡圓所在平面上建立平面直角座標 (x, y)，使得質點逆時針繞原點公轉，即此座標的 $(1,0)$ 與 $(0,1)$ 在三維空間中的叉積為角速度方向單位向量，令軌跡圓半徑 R，位置 **x**：

$$\mathbf{x} = \big(R\cos(\omega t+\phi), R\sin(\omega t+\phi)\big)$$

$$\dot{\mathbf{x}} = \boldsymbol{\omega}\times\mathbf{x} = \big(-R\omega\sin(\omega t+\phi), R\omega\cos(\omega t+\phi)\big)$$

$$\ddot{\mathbf{x}} = \boldsymbol{\omega}\times\dot{\mathbf{x}} = \big(-R\omega^2\cos(\omega t+\phi), -R\omega^2\sin(\omega t+\phi)\big)$$

其中初始相位角 ϕ，相位角 $\omega t+\phi$，速率 $R\omega$、加速度量值 $R\omega^2$。

xxxiv Gyroscopic precession (陀螺儀進動)

Suppose a top rotating at angular momentum **L** is subjected to a constant force **F** on a point on its rotational axis at position **r** relative to a fixed fulcrum, and another constant force −**F** on the fulcrum, which is also on its rotational axis. Then the tip of **L** is rotating at an angular velocity $\boldsymbol{\Omega}_p$ given by

$$\boldsymbol{\Omega}_p = \frac{\hat{\mathbf{L}}\times(\mathbf{r}\times\mathbf{F})}{|\mathbf{L}|}.$$

In special case that $\hat{\mathbf{L}} = \hat{\mathbf{r}}$, the tip of **L** is rotating at constant angular velocity around a precession axis fixed in the space, called steady precession.

xxxv Rotating reference frame

We derive the relation between the rotating reference frame R and the inertial reference frame I with the same origin where the former is rotating with respect to the later with angular velocity $\boldsymbol{\Omega}(t)$. We denote without subscript for frame I and subscript r for frame R.

Then for any unit vector of the rotating frame $\hat{\mathbf{u}}$:

$$\frac{\mathrm{d}}{\mathrm{d}t}\hat{\mathbf{u}} = \boldsymbol{\Omega} \times \hat{\mathbf{u}}$$

So, for a vector function **f**,

$$\frac{\mathrm{d}\mathbf{f}}{\mathrm{d}t} = \left(\frac{\mathrm{d}\mathbf{f}}{\mathrm{d}t}\right)_r + \left(\frac{\mathrm{d}\hat{\mathbf{i}}}{\mathrm{d}t}, \frac{\mathrm{d}\hat{\mathbf{j}}}{\mathrm{d}t}, \frac{\mathrm{d}\hat{\mathbf{k}}}{\mathrm{d}t}\right) \cdot \mathbf{f} = \left(\frac{\mathrm{d}\mathbf{f}}{\mathrm{d}t}\right)_r + \boldsymbol{\Omega} \times \mathbf{f}$$

Apply to velocity and acceleration

$$\mathbf{v} = \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t} = \mathbf{v}_r + \boldsymbol{\Omega} \times \mathbf{r}$$

$$\begin{aligned}
\mathbf{a} &= \frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} \\
&= \frac{\mathrm{d}\mathbf{v}_r}{\mathrm{d}t} + \frac{\mathrm{d}}{\mathrm{d}t}(\boldsymbol{\Omega} \times \mathbf{r}) \\
&= \left(\frac{\mathrm{d}\mathbf{v}_r}{\mathrm{d}t}\right)_r + \boldsymbol{\Omega} \times \mathbf{v}_r + \frac{\mathrm{d}\boldsymbol{\Omega}}{\mathrm{d}t} \times \mathbf{r} + \boldsymbol{\Omega} \times \mathbf{v} \\
&= \mathbf{a}_r + \boldsymbol{\Omega} \times \mathbf{v}_r + \frac{\mathrm{d}\boldsymbol{\Omega}}{\mathrm{d}t} \times \mathbf{r} + \boldsymbol{\Omega} \times \mathbf{v}_r + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}) \\
&= \mathbf{a}_r + 2\boldsymbol{\Omega} \times \mathbf{v}_r + \frac{\mathrm{d}\boldsymbol{\Omega}}{\mathrm{d}t} \times \mathbf{r} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r})
\end{aligned}$$

Thus, the fictitious force on a particle of mass m at **r** with velocity $\mathbf{v}_r$ in R is

$$-m\left(2\boldsymbol{\Omega} \times \mathbf{v}_r + \frac{\mathrm{d}\boldsymbol{\Omega}}{\mathrm{d}t} \times \mathbf{r} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r})\right),$$

where:

- $-2m\boldsymbol{\Omega} \times \mathbf{v}_r$ is called Coriolis force (科里奧利力/科氏力),
- $-m\frac{\mathrm{d}\boldsymbol{\Omega}}{\mathrm{d}t} \times \mathbf{r}$ is called Euler force (歐拉力), and
- $-m\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r})$ is called centrifugal force (離心力).

XI　摩擦力（Frictional force）

i　正向力（Normal Force）

垂直於兩系統界面的接觸排斥力，且令界面兩側系統為 A、B，界面指入 A 的單位法向量 $\hat{n}$，**A** 相對於 **B** 的速度 **v**，則當兩系統間正向力量值大於零時其必足夠小使得 $\mathbf{v}\cdot\hat{n}\leq 0$。源自於電磁力。

ii　摩擦力（Frictional force）

平行於兩系統界面的接觸力，且與受力系統相對於施力系統速度的內積不為正，且與受力系統所受其他外力合的內積不為正。與該界面的正向力與界面粗糙程度有關，與接觸面積無關。源自於電磁力。

iii　靜摩擦力（Static frictional force）

界面兩側系統相對速度平行界面的分量為零時的摩擦力，等於受力系統所受其他外力合平行界面的分量乘以負一，其量值不大於最大靜摩擦力。

iv　最大靜摩擦力（Maximum static frictional force）

一界面可能發生的靜摩擦力量值的最大值，其量值正比於該界面的正向力量值，其量值與正向力量值的比值稱靜摩擦（力）係數（Coefficient of static friction）μ_s。

v　（滑）動摩擦力（Kinetic frictional force）

界面兩側系統相對速度平行界面的分量不為零時的摩擦力，其量值與兩側系統相對速度平行界面的分量無關，其量值正比於該界面的正向力量值，其量值與正向力量值的比值稱（滑）動摩擦（力）係數（Coefficient of kinetic friction）μ_k，通常略小於靜摩擦（力）係數。

XII　Elasticity (彈性)

i　Introduction

Elasticity is the ability of a body to resist a distorting external force/stress (外力/外應力) and to return to its initial size and shape when that force/stress is removed. Such force and stress are called elastic force/stress, resisting force/stress, or restoring force/stress, and are the negation of the net external force and stress applied when the two achieves static equilibrium.

Linear elastic region or simply elastic region is a region of a type external stress in which the strain of the elastic object is proportional to stress, called linear elasticity. The bounds of the linear elastic region are called elastic limits (彈性限度) or yield strengths. If the stress continue to increase in magnitude, the object eventually ruptures, at a stress called the ultimate strength.

The restoring stress is only dependent on the strain (應變) of the object but not directly dependent on external stress in linear elastic region.

Four positive constants, called moduli of elasticity, are used to describe elasticity or stiffness of an elastic object within the elastic region, which are given by four stress-strain relations, collectively called generalized Hooke's law (虎克定律).

ii Young's modulus or Young modulus (楊氏模量/楊氏模數) (E) for axial elasticity

When an object is under uniaxial stress (stress perpendicular to the cross-section, i.e. tension or compression in one direction), the axial stress–strain relation during static equilibrium is given by

$$E = \frac{p}{\frac{\Delta L}{L}} = \frac{\frac{F}{A}}{\frac{\Delta L}{L}},$$

where $p = \frac{F}{A}$ is the external axial stress applied, postive when pointing outward the object and negative when pointing inward the object, F is the external axial force with same orientationas of p, A is the cross-sectional area, ΔL is the change of the axial length of the object, L is the initial axial length of the object, $\frac{\Delta L}{L}$ is called the axial strain of the object, and E is a positive constant called Young's modulus or Young modulus, which is only dependent on the material of the object.

The initial Hooke's law for linear spring is an implicate of the above relation, written

$$-F = -k(\Delta L),$$

where $-F$ is the restoring force, and k is a positive constant called force constant (力常數) or spring constant (彈力常數/彈性常數), which is given by

$$k = \frac{EA}{L}.$$

A spring that satisfies such relation is called an ideal spring (理想彈簧).

A potential energy U, called spring potential energy, can be defined, for the restoring force, which is conservative, as

$$U = \frac{k(\Delta L)^2}{2}.$$

iii Poisson's ratio (蒲松比/蒲松氏比) (ν) for transverse strain responding to uniaxial stress

When an object is under uniaxial stress (i.e. tension or compression in one direction), it tends to expand in directions perpendicular to the direction of compression, called Poisson's effect, in which the orthogonal stress–strain relation during static equilibrium is given by

$$\nu = -\frac{\frac{\Delta D}{D}}{\frac{\Delta L}{L}},$$

where ΔL is the change of the axial length of the object, L is the initial axial length of the object, $\frac{\Delta L}{L}$ is called the axial strain of the object, ΔD is the change of the transverse (any direction perpendicular to the axial direction) length of the object, D is the initial transverse (chosen to be the same direction of ΔD) length of the object, $\frac{\Delta D}{D}$ is called the transverse strain of the object, and ν is a positive constant called Poisson's ratio, which is only dependent on the material of the object.

iv Bulk modulus (體積模數/不可壓縮量) (B or K) for volumetric elasticity

When an object is under uniform stress (e.g. hydraulic compression due to stress exerted by a surrounding fluid), the stress-strain relation is given by

$$B = -\frac{\Delta p}{\frac{\Delta V}{V}},$$

where Δp is the change of external uniform pressure applied, ΔV is the chamge of the volume of the object, V is the initial volume of the object, $\frac{\Delta V}{V}$ is called the volume strain of the object, and B (sometimes K) is a positive constant called bulk modulus, which is only dependent on the material of the object.

v Shear modulus (剪切模量) or modulus of rigidity (G, S, μ) for shear elasticity

When an object is under tangential/shear stress (切/剪應力) (stress parallel to the cross-section), the stress–strain relation during static equilibrium is given by

$$G = \frac{\tau}{\gamma} = \frac{\frac{F}{A}}{\gamma} = \frac{\tau}{\frac{\Delta D}{L}} = \frac{\tau}{\tan\theta},$$

where $\tau = \frac{F}{A}$ is the external tangential stress applied, F is the external tangential force with same orientationas of p, A is the cross-sectional area, ΔD is the transverse displacement of the cross-section of the object in the direction F, L is the initial length perpendicular to the cross-section of the object, θ is the angle deformation of the object, $\gamma = \frac{\Delta D}{L} = \tan\theta$ is called the shear strain (剪應變) of the object, and G (sometimes S or μ) is a positive constant called shear modulus or modulus of rigidity, which is only dependent on the material of the object.

For small angle, the approximation

$$\gamma = \tan\theta \approx \theta$$

is sometimes used.

vi Relation among moduli

$$E = 3B(1 - 2\nu) = 2G(1 + \nu).$$

XIII　重力（Gravity）

i　牛頓萬有引力定律（Newton's Law of Universal Gravitation）

質量 M、m 的兩質點距離 r，則其相互重力量值 F 為：

$$F = \frac{GMm}{r^2},$$

方向指向另一質點，其中重力常數（Gravitational constant）$G = 6.67430 \times 10^{-11}$ $\mathrm{m^3\ kg^{-1}\ s^{-2}}$。

ii　重力場（Gravitational field）

一個質量 M 的質點發出的重力場 **g** 作為位置 **r** 的函數，即另一質點在該位置受到其重力產生之加速度，稱重力加速度（gravitational acceleration or acceleration of gravity），為：

$$\mathbf{g} = -\frac{GM\hat{\mathbf{r}}}{|\mathbf{r}|^2}$$

重力場強度（Gravitational field strength）為其量值。重力場為保守場。

iii　重力位能（Gravitational potential energy）

質量 M、m 的兩質點距離 r，其間儲存重力位能 U，以 $r = \infty$ 為 $U = 0$：

$$U = -\frac{GMm}{r}$$

iv　卡文迪西（扭秤）試驗（Cavendish experiment）

用線捆綁於中懸掛的長木棍兩端各掛一小鉛球作為扭秤，兩大鉛球分別懸掛在小球附近一小段距離，測量大球和小球之間微弱的重力。

v　球殼定理（Shell Theorem）

Theorem: 一半徑 R、球心 P 的球體系統 S，其總質量 M、質心位於 P、質量分布球對稱，則：

1. S 對與 P 距離 $\geq R$ 的任一點的淨重力效應，與一個質量 M、位於 P 的質點相同。
2. 令 S 內有一半徑 $r < R$、球心 P 的球體子系統 T，則 $S - T$ 對 T 內任一點的淨重力效應為零。

Proof. 將 S 視為由無限薄的薄球殼組成，則定理等同於：

一半徑 R、球心 P 的球殼系統 S，其總質量 m、質心位於 P、質量分布球對稱，則：

1. S 對與 P 距離 $\geq R$ 處的任一點的淨重力效應，與一個質量 m、位於 P 的質點相同。
2. S 對與 P 距離 $< R$ 處 的淨重力效應為零。

任選一通過 P 的平面，在其上定義一個以 P 為極點、任一在該平面上且始於 P 的射線為極軸的極座標系統。令重力場指向 P 為正、其反向為負。令球殼發出的重力場 g。將球殼視為由無限多個無限小的圓環組成，使得每個圓環的質心都在極軸上。定義每個圓環的角位置為其與極座標平面在極軸同一側的交點的角位置，使得每個圓環都有一個在 $[0, \pi]$ 的角位置。取極軸上一與 P 距離 D 的點 Q。令 $\mathrm{d}m$ 為角位置 θ 的圓環的質量、s 為該圓環上任意點與 Q 的距離、$\mathrm{d}g$ 為該圓環發出的重力場在 Q 的量值。令 $\mathrm{dd}m$ 為該圓環上任一點的質量、$\mathrm{dd}g$ 為該點發出的重力場在 Q 的量值。

令：

$$x = D - R\cos\theta$$

對於圓環上一點有：

$$\mathrm{dd}g = G\frac{\mathrm{dd}m}{s^2}$$

故對於任一圓環有：

$$\begin{aligned}\mathrm{d}g &= \int \mathrm{dd}g \\ &= \left(\frac{G}{s^2}\right)\int\left(\frac{(\mathrm{dd}m)x}{s}\right) \\ &= \frac{G(\mathrm{d}m)x}{s^3}\end{aligned}$$

對於角位置 θ 的圓環有：

$$R^2 + D^2 - s^2 = 2RD\cos\theta$$

$$x^2 = D^2 + R^2\cos^2\theta - 2RD\cos\theta = s^2 - R^2\sin^2\theta$$

故：

$$\begin{aligned}\mathrm{d}m &= \left(\frac{m}{4\pi R^2}\right)\cdot\left(R\,\mathrm{d}\theta\right)\cdot(2\pi\sqrt{s^2-x^2}) \\ &= \frac{m\,\mathrm{d}\theta\cdot\sqrt{s^2-x^2}}{2R} \\ &= \frac{m\,\mathrm{d}\theta\cdot\sin\theta}{2}\end{aligned}$$

故對於整個球殼有：

$$\begin{aligned}g &= \int \mathrm{d}g \\ &= \int_0^\pi\left(\frac{m\sin\theta}{2}\frac{Gx}{s^3}\right)\mathrm{d}\theta \\ &= \frac{Gm}{2}\int_0^\pi\left(\frac{\sin\theta\,(D - R\cos\theta)}{s^3}\right)\mathrm{d}\theta \\ &= \frac{Gm}{2}\int_0^\pi\left(\frac{\sin\theta\left(D - \frac{R^2+D^2-s^2}{2D}\right)}{s^3}\right)\mathrm{d}\theta \\ &= \frac{Gm}{2}\int_0^\pi\left(\frac{\sin\theta\left(D^2 - R^2 + s^2\right)}{2Ds^3}\right)\mathrm{d}\theta\end{aligned}$$

利用 $R^2 + D^2 - s^2 = 2RD\cos\theta$：

$$\mathrm{d}s\frac{\mathrm{d}\left(R^2 + D^2 - s^2\right)}{\mathrm{d}s} = \mathrm{d}\theta\frac{\mathrm{d}\left(2RD\cos\theta\right)}{\mathrm{d}\theta}$$

$$-\mathrm{d}s \cdot 2s = \mathrm{d}\theta \cdot (-2RD\sin\theta)$$

$$s\,\mathrm{d}s = RD\sin\theta\,\mathrm{d}\theta$$

故：

$$\begin{aligned}
g &= \frac{Gm}{2}\int_{|D-R|}^{D+R}\left(\frac{\sin\theta\left(D^2 - R^2 + s^2\right)}{2Ds^3}\frac{s}{RD\sin\theta}\right)\mathrm{d}s \\
&= \frac{Gm}{4RD^2}\int_{|D-R|}^{D+R}\left(\frac{D^2 - R^2}{s^2} + 1\right)\mathrm{d}s \\
&= \frac{Gm}{4RD^2}\left(\frac{R^2 - D^2}{s} + s\right)\Bigg|_{|D-R|}^{D+R} \\
&= \frac{Gm}{4RD^2}\left(\frac{R^2 - D^2}{D+R} - \frac{R^2 - D^2}{|D-R|} + D + R - |D-R|\right) \\
&= \begin{cases} \dfrac{Gm}{4RD^2}(R - D + R + D + D + R - D + R) = \dfrac{Gm}{D^2}, & D \geq R \\ \dfrac{Gm}{4RD^2}(R - D - R - D + D + R - R + D) = 0, & D < R \end{cases}
\end{aligned}$$

注意到 $D = R$ 當且僅當 $D = R = 0$，此時球殼為一質點。 □

vi 空腔定理（Cavity Theorem）

Theorem: 一半徑 R、球心 P 的球體 S，其總質量 M、質心位於 P、質量分布均勻。將 S 內挖出一半徑 r、球心 Q、質量為零的球體空腔 T，其中 $r \leq R - \overline{PQ}$。令一動點 W 位於 T 中，則該點之重力場 $\vec{g}$ 與 W 的位置無關。

Proof.

$$\vec{g} = \frac{GM\overrightarrow{WP}}{R^3} - \frac{GM\overrightarrow{WQ}}{R^3} = \frac{GM\overrightarrow{QP}}{R^3}$$

□

vii 標準重力加速度（Standard acceleration of gravity）

$$g_0 \coloneqq 9.80665\ \mathrm{m\ s^{-2}}$$

viii 重量（Weight）

物體所受的重力。

ix 視重（Apparent Weight）

一般定義為系統受到的外接觸力在其所受重力反方向的分量。

x 重心（Center of gravity）

若一質量 m 得系統受的重力與一位於點 P、質量 m 的質點受的重力相同，則稱點 P 為該系統的重心。

若系統各處重力場相同則系統重心即質心。

xi General Relativity, general theory of relativity, or Einstein's theory of gravity

Einstein pointed out that gravitation and acceleration are equivalent, called principle of equivalence, and explained gravitational effects in terms of a curvature of four-dimensional spacetime due to mass.

XIV 均勻有效重力場下之運動

i Gravitation and centripetal acceleration due to Earth's rotation

Suppose Earth is a sphere rotating at constant angular speed ω about axis passing its center with density being a function of the distance of a point from the center of Earth. Let there be a mass point static in Earth reference frame at latitude φ and distance R from the center of Earth rotating with Earth. And let the point be origin, the y-axis point toward the center of Earth, and x-axis be perpendicular latitude line and such that the northern pole is with positive x for points in northern hemisphere and the southern pole is with positive x for points in southern hemisphere. Let the magnitude of Earth's gravitational acceleration at the point be a_g.

Then, the total acceleration of the point in Earth reference frame is

$$(0, a_g) + \omega^2 R\cos\theta(\sin\theta, -\cos\theta) = \left(\omega^2 R\frac{\sin(2\theta)}{2}, a_g - \omega^2 R\cos^2\theta\right).$$

Thus, the effective gravitational acceleration downward increases as latitude increases and has an absolute minimum $a_g - \omega^2 R$ on equator and absoute maxima a_g on both poles.

ii 錐動擺/圓錐擺（Conical pendulum）

一質量 m 質點懸掛於一長 L 之繩下端，繩上端固定，與重力場方向夾銳角 θ，質點僅受量值 N 之繩張力與量值 mg 之重力，做角速率 ω、速率 v、軌跡圓半徑 $L\sin\theta$ 之等速率圓周運動：

$$N = mg\sec\theta$$

$$g\tan\theta = \omega^2 L\sin\theta$$

$$\omega = \sqrt{\frac{g\sec\theta}{L}}$$

$$v = \sqrt{gL\sin\theta\tan\theta}$$

iii 自由落體運動（Free fall motion）

均勻重力加速度量值 g 下，一平面直角座標使得 x 軸在地面上、重力加速度 $(0,-g)$，一高度不計物體初位置 $(0,h)$、$t=0$ 時以仰角 $\theta=\arctan\frac{v_y}{v_x}$，速率 $v=\sqrt{{v_x}^2+{v_y}^2}$ 斜拋/發射（projectile）而出，即初速度 $(v_x,v_y)=(v\sin\theta,v\cos\theta)$，落地（即 $y=0$）前位置 **r**：

$$\mathbf{r}=\left(v_xt,h+v_yt-\frac{g}{2}t^2\right)$$

$$\dot{r}=\left(v_x,v_y-gt\right)$$

若 $v_y>0$，則到達最高點時：

$$t=\frac{v_y}{g}$$

$$\mathbf{r}=\left(\frac{v_xv_y}{g},h+\frac{{v_y}^2}{2g}\right)$$

$$\dot{r}=\left(v_x,0\right)$$

落地前，在到達最高點前與後相同時間之速率相同、高度相同、鉛直速度分量相反，稱時間對稱性。

到達地面時：

$$t=\frac{v_y+\sqrt{{v_y}^2+2gh}}{g}=\frac{v\sin\theta+\sqrt{v^2\sin^2\theta+2gh}}{g}$$

$$\mathbf{r}=\left(\frac{v_x\left(v_y+\sqrt{{v_y}^2+2gh}\right)}{g},0\right)=\left(v\cos\theta\frac{v\sin\theta+\sqrt{v^2\sin^2\theta+2gh}}{g},0\right)$$

$$\dot{r}=\left(v_x,-\sqrt{{v_y}^2+2gh}\right)=\left(v\cos\theta,-\sqrt{v^2\sin^2\theta+2gh}\right)$$

此時 **r** 的 x 座標稱射程（range）。

落地時與地面夾角 θ_g 為：

$$\tan\theta_g=\frac{\sqrt{{v_y}^2+2gh}}{v_x}=\frac{\sqrt{v^2\sin^2\theta+2gh}}{v\cos\theta}$$

給定速率 v 下最大射程 R 為

$$R=\frac{v\sqrt{v^2+2gh}}{g}$$

發生於

$$\sin\theta=\frac{v}{\sqrt{2\left(v^2+gh\right)}}$$

$$\cos\theta=\frac{\sqrt{v^2+2gh}}{\sqrt{2\left(v^2+gh\right)}}$$

$$\tan\theta = \frac{v}{\sqrt{v^2+2gh}}$$

到達地面時：

$$\dot{r} = \left(\frac{v\sqrt{v^2+2gh}}{\sqrt{2\left(v^2+gh\right)}}, -\frac{v^2+2gh}{\sqrt{2\left(v^2+gh\right)}} \right)$$

$$\tan\theta_g = \frac{\sqrt{v^2+2gh}}{v}$$

與 θ 互為補角。

Proof.

$$R = \frac{v\cos\theta}{g}\left(v\sin\theta + \sqrt{v^2\sin^2\theta + 2gh}\right)$$

$$s \coloneqq \sin\theta, \quad \cos\theta = \sqrt{1-s^2}$$

$$R(s) = \frac{v\sqrt{1-s^2}}{g}\left(vs + \sqrt{v^2s^2+2gh}\right)$$

$$R'(s) = \frac{-vs}{g\sqrt{1-s^2}}\left(vs+\sqrt{v^2s^2+2gh}\right) + \frac{v\sqrt{1-s^2}}{g}\left(v + \frac{v^2s}{\sqrt{v^2s^2+2gh}}\right)$$

$$R'(s) = 0 \Rightarrow$$

$$s\left(vs\sqrt{v^2s^2+2gh} + v^2s^2 + 2gh\right) = \left(1-s^2\right)\left(v\sqrt{v^2s^2+2gh} + v^2s\right)$$

$$vs^2\sqrt{v^2s^2+2gh} + v^2s^3 + 2ghs = v\sqrt{v^2s^2+2gh} + v^2s - vs^2\sqrt{v^2s^2+2gh} - v^2s^3$$

$$2ghs - v^2s\left(1-2s^2\right) = v\left(1-2s^2\right)\sqrt{v^2s^2+2gh}$$

$$(2gh)^2s^2 + v^4s^2\left(1-2s^2\right)^2 - 4ghv^2s^2\left(1-2s^2\right) = v^4s^2\left(1-2s^2\right)^2 + 2ghv^2\left(1-2s^2\right)^2$$

$$2ghs^2 = v^2\left(1-2s^2\right)$$

$$s^2 = \frac{v^2}{2\left(v^2+gh\right)}$$

$$s = \pm\frac{v}{\sqrt{2\left(v^2+gh\right)}}$$

$$R = \frac{v\cos\theta}{g}\left(v\sin\theta + \sqrt{v^2\sin^2\theta + 2gh}\right)$$

$$R\left(\pm\frac{v}{\sqrt{2\left(v^2+gh\right)}}\right) = \frac{v}{g}\sqrt{\frac{v^2+2gh}{2\left(v^2+gh\right)}}\left(\pm\frac{v^2}{\sqrt{2\left(v^2+gh\right)}} + \sqrt{\frac{v^4}{2\left(v^2+gh\right)}+2gh}\right)$$

$$= \frac{v}{g}\frac{\sqrt{v^2+2gh}}{2\left(v^2+gh\right)}\left(\pm v^2 + v^2 + 2gh\right)$$

± 取正較大。

$$R\left(\frac{v}{\sqrt{2\left(v^2+gh\right)}}\right)=\frac{v}{g}\frac{\sqrt{v^2+2gh}}{2\left(v^2+gh\right)}\left(2v^2+2gh\right)$$
$$=\frac{v\sqrt{v^2+2gh}}{g}$$

□

若 $h=0$，則到達地面時：

$$t=\frac{2v_y}{g}$$
$$\mathbf{r}=\left(\frac{2v_xv_y}{g},0\right)$$
$$\dot{r}=\left(v_x,-v_y\right)$$

最大射程 R 為

$$R=\frac{v^2}{g}$$

發生於

$$\theta=\theta_g=\frac{\pi}{4}$$

iv　符合阻力方程式之阻力下的自由落體運動

均勻重力加速度量值 g 下，一平面直角座標使得 x 軸在地面上、重力加速度 $\mathbf{g}=(0,-g)$，設質量 m、速度 $\dot{r}$ 之物體所受空氣阻力 $\mathbf{f}$ 為：

$$\mathbf{f}=-\frac{1}{2}C\rho A\left|\dot{r}\right|^2\hat{r}$$

其中 $\frac{1}{2}C\rho A$ 為正常數。

則該物體之終端速度 v_t 為：

$$v_t=\sqrt{\frac{2g}{C\rho A}}.$$

v　符合斯托克定律之阻力下的自由落體運動

均勻重力加速度量值 g 下，一平面直角座標使得 x 軸在地面上、重力加速度 $\mathbf{g}=(0,-g)$，設質量 m、速度 $\dot{r}$ 之物體所受空氣阻力 $\mathbf{f}$ 為：

$$\mathbf{f}=-mb\dot{r}$$

其中 b 為正常數。

一高度不計物體初位置 $(0,h)$、$t=0$ 時以初速度 (v_x,v_y) 斜拋而出，落地（即 $y=0$）前位置 $\mathbf{r}$：

$$\ddot{\mathbf{r}}=-b\dot{\mathbf{r}}+\mathbf{g}$$

$$\mathbf{r} = \left(\frac{v_x}{b} \left(1 - e^{-bt} \right), h + \frac{gt}{b} - \left(\frac{v_y}{b} + \frac{g}{b^2} \right) \left(1 - e^{-bt} \right) \right)$$

$$\dot{\mathbf{r}} = \left(v_x e^{-bt}, \frac{g}{b} + \left(v_y + \frac{g}{b} \right) e^{-bt} \right)$$

$$\ddot{\mathbf{r}} = \left(-b v_x e^{-bt}, -\left(b v_y + g \right) e^{-bt} \right)$$

Proof.

$$(u, w) \coloneqq \dot{\mathbf{r}}$$

$$\dot{u} + bu = 0$$

$$u = v_x e^{-bt}$$

$$\dot{w} + bw = -g$$

$$e^{bt} \dot{w} + b e^{bt} w = -g e^{bt}$$

$$\frac{\mathrm{d}}{\mathrm{d}t} \left(e^{bt} w \right) = -g e^{bt}$$

$$e^{bt} w = -\frac{g}{b} e^{bt} + B$$

$$w = -\frac{g}{b} + B e^{-bt}$$

$$-\frac{g}{b} + B = v_y$$

$$w = \frac{g}{b} + \left(v_y + \frac{g}{b} \right) e^{-bt}$$

$$\dot{\mathbf{r}} = \left(v_x e^{-bt}, \frac{g}{b} + \left(v_y + \frac{g}{b} \right) e^{-bt} \right)$$

$$\ddot{\mathbf{r}} = \left(-b v_x e^{-bt}, -\left(b v_y + g \right) e^{-bt} \right)$$

$$\mathbf{r} = \left(\frac{v_x}{b} \left(1 - e^{-bt} \right), h + \frac{gt}{b} - \left(\frac{v_y}{b} + \frac{g}{b^2} \right) \left(1 - e^{-bt} \right) \right)$$

□

終端速度為：

$$\dot{\mathbf{r}} = \left(0, -\frac{g}{b} \right)$$

Proof.

$$-b\dot{\mathbf{r}} + (0, -g) = 0$$

$$\dot{\mathbf{r}} = \left(0, -\frac{g}{b} \right)$$

□

XV　忽略恆星運動的雙星重力運動

下星體均視為質點，行星泛指相對於恆星質量極小而可將後者視為不動之小天體。

i 克卜勒第一行星運動定律（Kepler's First Laws of Planetary Motion）/克卜勒橢圓定律（Kepler's Law of Ellipses）

星與恆星的系統，行星運動軌跡為以恆星為圓心的圓、以恆星為一焦點的橢圓、以恆星為焦點的拋物線、以恆星為距離該分支頂點較近的焦點的一雙曲線分支，或一通過恆星的直線的一部分。

Proof. 以恆星位置為原點，在行星軌跡與恆星所在的平面為建立極座標，令時間 t，恆星質量 M，行星質量 m，位置 $(r;\theta)$，$j \coloneqq GM$，萬有引力給出：

$$\ddot{r} - r\dot{\theta}^2 = -\frac{j}{r^2}$$

以恆星為參考點之角動量 $L = mr^2\dot{\theta}$ 守恆，令：

$$h \coloneqq \frac{L}{m} = r^2\dot{\theta}$$

若 $h = 0$，則行星運動軌跡為一通過恆星的直線的一部分。

討論 $h \neq 0$ 的情況，令 $u = \frac{1}{r}$：

$$2u^{-3}\dot{u}^2 - u^{-2}\ddot{u} - \frac{\dot{\theta}^2}{u} = -ju^2$$

變數轉換為 $u(\theta)$：

$$\dot{\theta} = hu^2$$

$$\dot{u} = \frac{\mathrm{d}u}{\mathrm{d}\theta}\dot{\theta} = hu^2\frac{\mathrm{d}u}{\mathrm{d}\theta}$$

$$\ddot{\theta} = 2hu\dot{u} = 2h^2u^3\frac{\mathrm{d}u}{\mathrm{d}\theta}$$

$$\ddot{u} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\mathrm{d}u}{\mathrm{d}\theta}\dot{\theta}\right) = \frac{\mathrm{d}^2u}{\mathrm{d}\theta^2}\dot{\theta}^2 + \frac{\mathrm{d}u}{\mathrm{d}\theta}\ddot{\theta} = h^2u^4\frac{\mathrm{d}^2u}{\mathrm{d}\theta^2} + 2h^2u^3\left(\frac{\mathrm{d}u}{\mathrm{d}\theta}\right)^2$$

$$2h^2u\left(\frac{\mathrm{d}u}{\mathrm{d}\theta}\right)^2 - h^2u^2\frac{\mathrm{d}^2u}{\mathrm{d}\theta^2} - 2h^2u\left(\frac{\mathrm{d}u}{\mathrm{d}\theta}\right)^2 - h^2u^3 = -ju^2$$

$$\frac{\mathrm{d}^2u}{\mathrm{d}\theta^2} + u = \frac{j}{h^2}$$

解：

$$\lambda^2 + 1 = 0$$

$$\lambda = \pm i$$

$$u = C\cos(\theta + \phi) + \frac{j}{h^2}$$

其中 C 和 ϕ 為常數。

$$
\begin{aligned}
r &= \frac{1}{C\cos(\theta+\phi)+\frac{j}{h^2}} \\
&= \frac{\frac{h^2}{j}}{1+\frac{Ch^2}{j}\cos(\theta+\phi)} \\
&= \frac{\frac{L^2}{GMm^2}}{1+\frac{CL^2}{GMm^2}\cos(\theta+\phi)}
\end{aligned}
$$

即：行星軌跡為，以 $\ell \coloneqq \frac{L^2}{GMm^2}$ 為半正焦弦（semi-latus rectum）、以 $\varepsilon \coloneqq \frac{CL^2}{GMm^2}$ 為離心率（Eccentricity）的，以恆星為圓心的圓、以恆星為一焦點的橢圓、以恆星為焦點的拋物線、以恆星為距離該分支頂點較近的焦點的一雙曲線分支。 □

ii 克卜勒第二行星運動定律（Kepler's Second Laws of Planetary Motion）/克卜勒等面積定律（Kepler's Law of Equal Areas）

行星與恆星的系統，行星與恆星連線單位時間掃過的面積恆不變。

Proof. 即 $\frac{h}{2} = \frac{L}{2m}$，因角動量守恆而不變。 □

iii 克卜勒第三行星運動定律（Kepler's Third Law of Planetary Motion）/克卜勒週期定律（Kepler's Law of Period）

忽略所有行星對其他行星的重力效應，繞同一恆星公轉的所有行星其橢圓軌道半長軸或圓軌道半徑 a 的三次方與週期 T 的二次方正比。

$$
\frac{a^3}{T^2} = \frac{GM}{4\pi^2}
$$

Proof. 令橢圓軌道半短軸或圓軌道半徑 b，週期 T 等於行星軌道面積 πab 除以行星與恆星連線單位時間掃過的面積 $\frac{j}{2}$。

$$
\begin{aligned}
b &= \sqrt{a\ell} \\
T &= \frac{2\pi ab}{j} \\
\ell &= \frac{L^2}{GMm^2} \\
L &= m\sqrt{GM\ell} \\
j &= \frac{\sqrt{GM\ell}}{2} \\
\frac{a^3}{T^2} &= \frac{a^3 j^2}{4\pi^2 a^2 b^2} \\
&= \frac{a^3 j^2}{4\pi^2 a^3 \ell} \\
&= \frac{GM}{4\pi^2}
\end{aligned}
$$

□

iv　行星公轉

對於橢圓軌道，距恆星較近的長軸頂點稱**近恆星點（perihelion）**，距恆星較遠的長軸頂點稱**遠恆星點（aphelion）**。一質量 m 行星以半長軸 a、焦距 c 橢圓或半徑 a 圓軌道繞質量 M 恆星運動，軌道離心率 ε，兩者距離 r，重力位能 U 以 $r=\infty$ 為 $U=0$，相對恆星的慣性參考系中，行星速率 v，動能 K，力學能 E 稱束縛能，角動量量值 L，對於橢圓軌道，行星位於近恆星點時距恆星 r_1、速率 v_1，行星位於遠恆星點時距恆星 r_2、速率 v_2，行星位於短軸頂點（即與恆星距離 $\sqrt{a^2-c^2}$ 處）時速率 v_b；對於圓軌道，$r_1=r_2=r=a$，行星速率 $v_1=v_2=v_b=v$：

$$r_1=a-c$$

$$r_2=a+c$$

$$\varepsilon=\frac{a}{c}$$

$$U=-\frac{GMm}{r}$$

$$E=-\frac{GMm}{2a}$$

$$K=GMm\left(\frac{1}{r}-\frac{1}{2a}\right)$$

$$L=m\sqrt{\frac{GM(a+c)(a-c)}{a}}=m\sqrt{GMa(1-\varepsilon^2)}$$

$$v=\sqrt{GM\left(\frac{2}{r}-\frac{1}{a}\right)}$$

$$v_1=\sqrt{\frac{GM(a+c)}{a(a-c)}}$$

$$v_2=\sqrt{\frac{GM(a-c)}{a(a+c)}}$$

$$v_b=\sqrt{\frac{GM}{a}}$$

Proof.

$$E=\frac{m{v_1}^2}{2}-\frac{GMm}{r_1}=\frac{m{v_2}^2}{2}-\frac{GMm}{r_2}$$

$${v_1}^2-\frac{2GM(a+c)}{a^2-c^2}={v_2}^2-\frac{2GM(a-c)}{a^2-c^2}$$

$$u\coloneqq\frac{v_1}{r_2}=\frac{v_2}{r_1}$$

$$u^2\left((a+c)^2-(a-c)^2\right)=\frac{4GMc}{a^2-c^2}$$

$$u=\sqrt{\frac{GM}{a(a+c)(a-c)}}$$

$$\frac{L}{m} = r_1 r_2 u = \sqrt{\frac{GM(a+c)(a-c)}{a}}$$

$$v_1 = ur_2 = \sqrt{\frac{GM(a+c)}{a(a-c)}}$$

$$v_2 = ur_1 = \sqrt{\frac{GM(a-c)}{a(a+c)}}$$

$$v_b = \frac{L}{m\sqrt{a^2-c^2}} = \sqrt{\frac{GM}{a}}$$

$$E = \frac{GMm(a+c)}{2a(a-c)} - \frac{GMm}{a-c} = \frac{GMm(a+c) - 2GMma}{2a(a-c)} = -\frac{GMm}{2a}$$

$$K = E - U = \frac{GMm}{r} - \frac{GMm}{2a}$$

$$v = \sqrt{\frac{2K}{m}} = \sqrt{GM\left(\frac{2}{r} - \frac{1}{a}\right)}$$

□

v　行星圓軌道公轉

一質量 m 行星以半徑 r 圓軌道繞質量 M 恆星運動，重力位能 U 以 $r = \infty$ 為 $U = 0$，相對恆星的慣性參考系中，行星速率 v，角速率 ω，動能 K，力學能 E 稱束縛能，角動量量值 L：

$$U = -\frac{GMm}{r}$$

$$E = -\frac{GMm}{2r}$$

$$K = \frac{GMm}{2r}$$

$$L = m\sqrt{GMr}$$

$$v = \sqrt{\frac{GM}{r}}$$

$$\omega = \sqrt{\frac{GM}{r^3}}$$

vi　行星掠過恆星

一質量 m 行星自無限遠處以速率 u 靠近質量 M 恆星，在無限遠處過其位置且以其速度為方向向量的直線與恆星距離 b，兩者距離 r，重力位能 U 以 $r = \infty$ 為 $U = 0$，相對恆星的慣性參考系中，行星速率 v，動能 K，力學能 $E = U + K$，角動量量值 L，與恆星最近距離 R，此時速率 w：

$$U = -\frac{GMm}{r}$$

$$E = \frac{mu^2}{2}$$

$$K = \frac{mu^2}{2} + \frac{GMm}{r}$$

$$\frac{L}{m} = ub = Rw$$

$$v = \sqrt{u^2 + \frac{2GM}{r}}$$

$$w = \frac{\sqrt{G^2M^2 + u^4b^2} + GM}{ub}$$

$$R = \frac{\sqrt{G^2M^2 + u^4b^2} - GM}{u^2}$$

Proof.

$$E = \frac{mv^2}{2} - \frac{GMm}{r} = \frac{mu^2}{2}$$

$$v^2 - u^2 = \frac{2GM}{r}$$

$$v = \sqrt{u^2 + \frac{2GM}{r}}$$

$$\frac{L}{m} = ub = Rw$$

$$R = \frac{ub}{w}$$

$$w^2 - \frac{2GMw}{ub} - u^2 = 0$$

$$w = \frac{GM}{ub} + \sqrt{\left(\frac{GM}{ub}\right)^2 + u^2} = \frac{GM + \sqrt{G^2M^2 + u^4b^2}}{ub}$$

$$R = \frac{u^2b^2}{GM + \sqrt{G^2M^2 + u^4b^2}} = \frac{\sqrt{G^2M^2 + u^4b^2} - GM}{u^2}$$

□

vii 行星公轉橢圓軌道的離心率向量（eccentricity vector）、真近點角（True anomaly）、偏近點角（Eccentricity anomaly）與平近點角（Mean anomaly）

離心率向量 ε 定義為量值等於**離心率** ε、方向為焦點指向近恆星點的向量。行星位置 $\mathbf{r}$ 以恆星 S 為原點，**真近點角** ν 定義為：

$$\nu = \arccos \frac{\boldsymbol{\varepsilon} \cdot \mathbf{r}}{|\boldsymbol{\varepsilon}||\mathbf{r}|}.$$

半長軸 a，**偏近點角** $\mathscr{E}$ 定義為使得近恆星點為 $\mathscr{E} = 0 + 2z\pi, \quad z \in \mathbb{Z}$ 且逆時針旋轉 $\mathscr{E}$ 增加且：

$$\cos \mathscr{E} = \frac{a - |\mathbf{r}|}{a\varepsilon}.$$

平近點角 $\mathscr{M}$ 定義為：

$$\mathscr{M} = \mathscr{E} - \varepsilon \sin \mathscr{E}.$$

令半短軸 $b = a\sqrt{1-\varepsilon^2}$，以 S 為原點、近日點在 x 軸正向上，建立平面直角座標，軌道上任一點 P 的位置 (x, y)，則：

$$x = a(\cos \mathscr{E} - \varepsilon)$$

$$y = b \sin \mathscr{E}$$

Proof. 以橢圓中心 C 為圓心、a 為半徑畫輔助圓，將 P 對應到輔助圓上一點 Q 使得直線 PQ 垂直橢圓長軸於 W。建立極座標 $(r; \theta) = \big(r\cos\theta, r\sin\theta\big)$。

$$\cos \angle WCQ = \frac{a\varepsilon + r\cos\theta}{a}$$

$$r = \frac{a(1-\varepsilon^2)}{1 + \varepsilon\cos(\theta)}$$

$$\cos\theta = \frac{a(1-\varepsilon^2) - r}{r\varepsilon}$$

$$\begin{aligned}\cos \angle WCQ &= \frac{a\varepsilon + \frac{a(1-\varepsilon^2)-r}{\varepsilon}}{a} \\ &= \frac{a\varepsilon^2 + a(1-\varepsilon^2) - r}{a\varepsilon} \\ &= \frac{a-r}{a\varepsilon}\end{aligned}$$

故：

$$\angle WCQ = \mathscr{E}$$

$$x = a(\cos \mathscr{E} - \varepsilon)$$

以橢圓中心 C 為圓心、b 為半徑畫輔助圓，將 P 對應到輔助圓上一點 F 使得直線 PF 垂直橢圓短軸於 X，因為：

$$\angle QPF = \angle QWC = \frac{\pi}{2}$$

$$\angle PQF = \angle WQC$$

故：

$$\angle CFX = \angle QFP = \angle WCQ = \mathscr{E}$$

$$y = b \sin \mathscr{E}$$

□

行星與恆星連線自近恆星點起掃過的面積 A，週期 T：

$$A = \frac{ab\mathscr{M}}{2}$$

$$\frac{\mathrm{d}\mathscr{M}}{\mathrm{d}t} = \frac{2\pi}{T}$$

Proof.

$$\begin{aligned}
A &= \int_0^{\mathscr{E}} \frac{1}{2}\left(x\frac{\mathrm{d}y}{\mathrm{d}\mathscr{E}} - y\frac{\mathrm{d}x}{\mathrm{d}\mathscr{E}}\right)\,\mathrm{d}\mathscr{E} \\
&= \int_0^{\mathscr{E}} \frac{1}{2}\left(a(\cos z-\varepsilon)b\cos z + b\sin z a\sin z\right)\,\mathrm{d}z \\
&= \int_0^{\mathscr{E}} \frac{1}{2}\left(ab - ab\varepsilon\cos z\right)\,\mathrm{d}z \\
&= \frac{ab}{2}\left(z-\varepsilon\sin z\right)\Big|_0^{\mathscr{E}} \\
&= \frac{ab}{2}\left(\mathscr{E}-\varepsilon\sin\mathscr{E}\right) \\
&= \frac{ab\mathscr{M}}{2}
\end{aligned}$$

□

一行星以離心率 ε、橢圓半長軸或圓半徑 a 軌道繞質量 M 恆星運動，週期 T，從一個短軸頂點經過近恆星點到達另一個短軸頂點花費時間 T_1，從一個短軸頂點經過遠恆星點到達另一個短軸頂點花費時間 T_2：

$$T = 2\pi\sqrt{\frac{a^3}{GM}}$$

$$T_1 = T\left(\frac{1}{2}-\frac{\varepsilon}{\pi}\right) = \sqrt{\frac{a^3}{GM}}\,(\pi-2\varepsilon)$$

$$T_2 = T\left(\frac{1}{2}+\frac{\varepsilon}{\pi}\right) = \sqrt{\frac{a^3}{GM}}\,(\pi+2\varepsilon)$$

viii　拉普拉斯-龍格-冷次向量（Laplace–Runge–Lenz vector, LRL vector）

令一質點質量 m、位置 **r**、受力 **F**：

$$\mathbf{F} = -\frac{k\hat{\mathbf{r}}}{|\mathbf{r}|^2}$$

、動量 **p**、角動量 $\mathbf{L} = \mathbf{r}\times\mathbf{p}$，則其拉普拉斯-龍格-冷次向量 **A** 被定義為：

$$\mathbf{A} = \mathbf{p}\times\mathbf{L} - mk\hat{\mathbf{r}}.$$

例如，在前述忽略恆星運動的雙星重力運動中，恆星為原點，$k = GMm$，行星有拉普拉斯-龍格-冷次向量 **A**：

$$\mathbf{A} = \mathbf{p}\times\mathbf{L} - GMm^2\hat{\mathbf{r}}.$$

拉普拉斯-龍格-冷次向量具有守恆性：

$$\frac{\mathrm{d}\mathbf{A}}{\mathrm{d}t} = 0.$$

Proof.

$$\frac{\mathrm{d}\mathbf{A}}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\mathbf{p}\times\mathbf{L}\right) - \frac{\mathrm{d}}{\mathrm{d}t}\left(mk\hat{\mathbf{r}}\right)$$

由於角動量守恆，$\frac{\mathrm{d}}{\mathrm{d}t}\left(\mathbf{p}\times\mathbf{L}\right)=\dot{\mathbf{p}}\mathbf{L}$。

$$\begin{aligned}\frac{\mathrm{d}\mathbf{A}}{\mathrm{d}t}&=\dot{\mathbf{p}}\mathbf{L}-mk\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{L\times\hat{\mathbf{r}}}{m\left|\mathbf{r}\right|^2}\right)\\&=-\frac{k\hat{\mathbf{r}}\times\mathbf{L}}{\left|\mathbf{r}\right|^2}-\frac{k\mathbf{L}\times\hat{\mathbf{r}}}{\left|\mathbf{r}\right|^2}\\&=0\end{aligned}$$

□

拉普拉斯-龍格-冷次向量與位置點積有：

$$\mathbf{A}\cdot\mathbf{r}=L^2-mk\left|\mathbf{r}\right|.$$

力學能為：

$$E=\frac{\left|\mathbf{p}\right|^2}{2m}-\frac{k}{r}.$$

拉普拉斯-龍格-冷次向量與自身點積有：

$$\left|\mathbf{A}\right|^2=2mE\left|\mathbf{L}\right|^2+m^2k^2.$$

Proof.

$$\begin{aligned}\left|\mathbf{A}\right|^2&=\left|\mathbf{p}\right|^2\left|\mathbf{L}\right|^2-2mk\frac{\left|\mathbf{L}\right|^2}{\left|\mathbf{r}\right|}+m^2k^2\\&=2mE\left|\mathbf{L}\right|^2+m^2k^2\end{aligned}$$

□

拉普拉斯-龍格-冷次向量與離心率 ε 的關係為：

$$\varepsilon=\frac{\left|\mathbf{A}\right|}{mk}.$$

Proof. 使用克卜勒第一行星運動定律的結果，離心率 ε 為：

$$\varepsilon=\frac{CL^2}{mk}$$

與半正焦弦為：

$$\ell=\frac{L^2}{mk}$$

又半長軸 a 為：

$$\begin{aligned}a&=\frac{\ell}{1-\varepsilon^2}\\&=\frac{L^2}{mk\left(1-\frac{C^2L^4}{m^2k^2}\right)}\\1-\frac{C^2L^4}{m^2k^2}&=\frac{L^2}{mka}\end{aligned}$$

$$\varepsilon^2 = \frac{C^2L^4}{m^2k^2} = 1 - \frac{L^2}{mka}.$$

又：

$$\begin{aligned} E &= -\frac{k}{2a} \\ \left(\frac{|\mathbf{A}|}{mk}\right)^2 &= 1 + \frac{2E\,|\mathbf{L}|^2}{mk^2} \\ &= 1 - \frac{|\mathbf{L}|^2}{mka} \\ &= \varepsilon^2 \end{aligned}$$

□

ix　逃逸/脫離速度（Escape velocity）與宇宙速度（Cosmic velocity）

令地球半徑 R、地表重力場量值 g。

- **逃逸/脫離速度（Escape velocity）**：以無限遠處為所有位能為零處，在某處使得力學能為零的速率為其逃逸/脫離速度。
- **第一宇宙速度（First cosmic velocity）/環繞速度**：在地球上發射的物體在地球表面作圓周運動所需的最小初始速度，即 $\sqrt{gR} \approx 7.9$ km/s。
- **第二宇宙速度（Second cosmic velocity）/地球逃逸/脫離速度**：只考慮與地球間的重力位能而不考慮與其他物質間的重力位能下地表的逃逸速度，即 $\sqrt{2gR} \approx 11.2$ km/s。
- **第三宇宙速度（Third cosmic velocity）/太陽系逃逸/脫離速度**：只考慮與地球間和與太陽間的重力位能而不考慮與其他物質間的重力位能下地表的逃逸速度，≈ 16.7 km/s。
- **第四宇宙速度（Fourth cosmic velocity）/銀河系逃逸/脫離速度**：只考慮與地球間、與太陽間和與銀河系間的重力位能而不考慮與其他物質間的重力位能下地表的逃逸速度，目前粗估在 525 km/s 以上。

x　地球同步軌道（Geosynchronous orbit, GSO）

在地球上空使得繞地球公轉週期與地球自轉週期相同的衛星軌道，海拔高度約 35786 km，其中衛星稱同步衛星（Geosynchronous satellite）。特別地，赤道上空的地球同步軌道稱地球靜止軌道（Geostationary orbit, GEO），其上衛星始終在地球同一地上空。

xi　內行星最大視角

設行星 A 以半徑 R 的圓軌道繞恆星 S 公轉，同平面上其內行星 B 以半徑 r 的圓軌道繞恆星 S 公轉，則 A 觀察 B 與 S 之夾角最大值為 $\arcsin\frac{r}{R}$，發生於 $\overline{AB} \perp \overline{BS}$ 時。

XVI　雙星重力運動

i　克卜勒第一行星運動定律（Kepler's First Laws of Planetary Motion）/克卜勒橢圓定律（Kepler's Law of Ellipses）

雙星系統，兩星之運動軌跡各為以雙星質心為圓心的圓、以雙星質心為一焦點的橢圓、以雙星質心為焦點的拋物線、以雙星質心為距離該分支頂點較近的焦點的一雙曲線分支，或一通過雙星質心的直線的一部分，且兩星之運動軌跡離心率相同、半正焦弦與星之質量反比。

Proof. 以雙星質心位置為原點，在雙星所在的平面為建立極座標，令時間 t，雙星質量 m、M，位置 $(r;\theta)$、$(R;\tau)$，雙星質心位置給出：

$$mr = MR$$

$$\tau = \theta \pm \pi$$

萬有引力給出：

$$\ddot{r} - r\dot{\theta}^2 = -\frac{GM}{(r+R)^2}$$

$$\ddot{R} - R\dot{\theta}^2 = -\frac{Gm}{(r+R)^2}$$

以雙星質心為參考點之角動量 $I = mr^2\dot{\theta}$ 與 $J = MR^2\dot{\theta}$ 守恆，令：

$$h \coloneqq \frac{I}{m} = r^2\dot{\theta}$$

$$H \coloneqq \frac{J}{M} = R^2\dot{\theta}$$

因為：

$$h = \frac{Hr^2}{R^2} = \frac{HM^2}{m^2}$$

$h = 0 \iff H = 0$，此時雙星運動軌跡各為一通過質心的直線的一部分。

討論 $h \neq 0$ 的情況，令 $u = \frac{1}{r}$，$w = \frac{1}{R}$：

$$\frac{1}{R+r} = \frac{uw}{u+w}$$

$$2u^{-3}\dot{u}^2 - u^{-2}\ddot{u} - \frac{\dot{\theta}^2}{u} = -GM\left(\frac{uw}{u+w}\right)^2$$

變數轉換為 $u(\theta)$：

$$\dot{\theta} = hu^2$$

$$\dot{u} = \frac{\mathrm{d}u}{\mathrm{d}\theta}\dot{\theta} = hu^2\frac{\mathrm{d}u}{\mathrm{d}\theta}$$

$$\ddot{\theta} = 2hu\dot{u} = 2h^2u^3\frac{\mathrm{d}u}{\mathrm{d}\theta}$$

$$\ddot{u} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\mathrm{d}u}{\mathrm{d}\theta}\dot{\theta}\right) = \frac{\mathrm{d}^2u}{\mathrm{d}\theta^2}\dot{\theta}^2 + \frac{\mathrm{d}u}{\mathrm{d}\theta}\ddot{\theta} = h^2u^4\frac{\mathrm{d}^2u}{\mathrm{d}\theta^2} + 2h^2u^3\left(\frac{\mathrm{d}u}{\mathrm{d}\theta}\right)^2$$

$$2h^2u\left(\frac{\mathrm{d}u}{\mathrm{d}\theta}\right)^2 - h^2u^2\frac{\mathrm{d}^2u}{\mathrm{d}\theta^2} - 2h^2u\left(\frac{\mathrm{d}u}{\mathrm{d}\theta}\right)^2 - h^2u^3 = -GM\left(\frac{uw}{u+w}\right)^2$$

$$h^2u^2\frac{\mathrm{d}^2u}{\mathrm{d}\theta^2} + h^2u^3 = GM\left(\frac{uw}{u+w}\right)^2$$

$$\frac{\mathrm{d}^2u}{\mathrm{d}\theta^2} + u = \frac{GM}{h^2}\left(\frac{w}{u+w}\right)^2$$

考慮：

$$w = \frac{Mu}{m}$$

令：

$$k \coloneqq \frac{\mathrm{d}^2u}{\mathrm{d}\theta^2} + u = \frac{GM}{h^2}\left(\frac{M}{m+M}\right)^2 = \frac{GM^3}{(m+M)^2h^2}$$

解：

$$\lambda^2 + 1 = 0$$

$$\lambda = \pm i$$

$$u = C\cos(\theta+\phi) + k$$

其中 C 和 ϕ 為常數。

$$\begin{aligned}
r &= \frac{1}{C\cos(\theta+\phi)+k} \\
&= \frac{\frac{1}{k}}{1+\frac{C}{k}\cos(\theta+\phi)} \\
&= \frac{\frac{1}{k}}{1+\frac{C}{k}\cos(\theta+\phi)} \\
&= \frac{\frac{(m+M)^2h^2}{GM^3}}{1+\frac{C(m+M)^2h^2}{GM^3}\cos(\theta+\phi)} \\
&= \frac{\frac{(m+M)^2I^2}{Gm^2M^3}}{1+\frac{C(m+M)^2I^2}{Gm^2M^3}\cos(\theta+\phi)}
\end{aligned}$$

令：

$$D \coloneqq CM$$

令雙星總角動量量值 L：

$$I = \frac{ML}{m+M}, \quad J = \frac{mL}{m+M}$$

有：

$$r = \frac{\frac{L^2}{Gm^2M}}{1+\frac{DL^2}{Gm^2M^2}\cos(\theta+\phi)}$$

又：

$$\begin{aligned}R&=\frac{mr}{M}\\&=\frac{\frac{m(m+M)^2h^2}{GM^4}}{1+\frac{C(m+M)^2h^2}{GM^3}\cos(\theta+\phi)}\\&=\frac{\frac{(m+M)^2H^2}{Gm^3}}{1-\frac{D(m+M)^2H^2}{Gm^4}\cos(\tau+\phi)}\\&=\frac{\frac{(m+M)^2J^2}{GM^2m^3}}{1-\frac{D(m+M)^2J^2}{GM^2m^4}\cos(\tau+\phi)}\\&=\frac{\frac{L^2}{GM^2m}}{1-\frac{DL^2}{GM^2m^2}\cos(\tau+\phi)}\end{aligned}$$

即：

- 質量 m 者軌跡為，以 $\ell_m\coloneqq\frac{L^2}{Gm^2M}$ 為半正焦弦、以 $\varepsilon\coloneqq\frac{DL^2}{Gm^2M^2}$ 為離心率的，以質心為圓心的圓、以質心為一焦點的橢圓、以質心為焦點的拋物線、以質心為距離該分支頂點較近的焦點的一雙曲線分支。
- 質量 M 者軌跡為，以 $\ell_M\coloneqq\frac{L^2}{GM^2m}$ 為半正焦弦、以 $\varepsilon\coloneqq\frac{DL^2}{GM^2m^2}$ 為離心率的，以質心為圓心的圓、以質心為一焦點的橢圓、以質心為焦點的拋物線、以質心為距離該分支頂點較近的焦點的一雙曲線分支。

□

ii 克卜勒第二行星運動定律（Kepler's Second Laws of Planetary Motion）/克卜勒等面積定律（Kepler's Law of Equal Areas）

雙星系統，一星與雙星質心連線單位時間掃過的面積恆不變，雙星連線單位時間掃過的面積亦恆不變。

Proof. 即 $\frac{h}{2}=\frac{I}{2m}$、$\frac{H}{2}=\frac{J}{2M}$ 與 $\frac{h+H}{2}=\frac{L|m-M|}{2mM}$，均因角動量守恆而不變。 □

iii 克卜勒第三行星運動定律（Kepler's Third Law of Planetary Motion）/克卜勒週期定律（Kepler's Law of Period）

質量 m、M 雙星互繞，兩者之橢圓軌道半長軸或圓軌道半徑 a、q、週期 T，則：

$$\frac{a^3}{T^2}=\frac{GM^3}{(m+M)^24\pi^2}$$

$$\frac{q^3}{T^2}=\frac{Gm^3}{(m+M)^24\pi^2}$$

$$\frac{(a+q)^3}{T^2}=\frac{G(m+M)}{4\pi^2}$$

Proof. 令橢圓軌道半短軸或圓軌道半徑 b，週期 T 等於該星軌道面積 πab 除以該星與雙星質心連線單位時間掃過的面積 $\frac{h}{2}$。

$$b = \sqrt{a\ell_m}$$

$$T = \frac{2\pi ab}{h}$$

$$\ell_m = \frac{L^2}{Gm^2M}$$

$$L = m\sqrt{GM\ell_m}$$

$$h = \frac{ML}{m(m+M)} = \frac{M\sqrt{GM\ell_m}}{m+M}$$

$$\begin{aligned}\frac{a^3}{T^2} &= \frac{a^3h^2}{4\pi^2a^2b^2} \\ &= \frac{a^3h^2}{4\pi^2a^3\ell_m} \\ &= \frac{GM^3}{(m+M)^2 4\pi^2}\end{aligned}$$

同理：

$$\frac{q^3}{T^2} = \frac{Gm^3}{(m+M)^2 4\pi^2}$$

$$a = M\left(\frac{Gm}{(m+M)^2 4\pi^2}\right)^{1/3}$$

$$q = m\left(\frac{Gm}{(m+M)^2 4\pi^2}\right)^{1/3}$$

$$a + q = (M+m)^3 \frac{Gm}{(m+M)^2 4\pi^2}$$

$$\frac{(a+q)^3}{T^2} = \frac{G(m+M)}{4\pi^2}$$

□

iv 雙星互繞

對於橢圓軌道，距質心較近的長軸頂點稱近質心點，距質心較遠的長軸頂點稱遠質心點，一星在近質心點時另一星亦在近質心點。質量 m、M 雙星分別以半長軸 $M\alpha$、$m\alpha$、焦距 $M\gamma$、$m\gamma$ 橢圓或半徑 $M\alpha$、$m\alpha$ 圓軌道繞雙星質心運動，軌道離心率 ε，兩者距離 r，分別距質心 $\frac{Mr}{m+M}$、$\frac{mr}{m+M}$，重力位能 U 以 $r = \infty$ 為 $U = 0$，相對雙星質心的慣性參考系中，兩者速率 $M\sigma$、$m\sigma$，總動能 $K = \frac{mM(m+M)\sigma^2}{2}$，總力學能 $E = U + K$ 稱束縛能，總角動量量值 L，對於橢圓軌道，雙星位於近質心點時雙星距離 r_1、兩者速率 $M\sigma_1$、$m\sigma_1$，雙星位於遠質心點時雙星距離 r_2、兩者速率 $M\sigma_2$、$m\sigma_2$，雙星位於短軸頂點時（即兩者與質心距離 $M\sqrt{\alpha^2-\gamma^2}$、$m\sqrt{\alpha^2-\gamma^2}$ 時）兩者速率 $M\sigma_b$、$m\sigma_b$；對於圓軌道，$r_1 = r_2 = r = (m+M)\alpha$，兩者速率 $M\sigma_1 = M\sigma_2 = M\sigma_b = M\sigma$、$m\sigma_1 = m\sigma_2 = m\sigma_b = m\sigma$：

$$r_1 = (m+M)(\alpha - \gamma)$$

$$r_2 = (m+M)(\alpha+\gamma)$$

$$\varepsilon = \frac{\alpha}{\gamma}$$

$$U = -\frac{GmM}{r}$$

$$E = -\frac{GmM}{2(m+M)\alpha}$$

$$K = GmM\left(\frac{1}{r} - \frac{1}{2(m+M)\alpha}\right)$$

$$L = mM\sqrt{\frac{G(\alpha+\gamma)(\alpha-\gamma)}{\alpha}}$$

$$\sigma = \sqrt{\frac{2G}{m+M}\left(\frac{1}{r} - \frac{1}{2(m+M)\alpha}\right)}$$

$$\sigma_1 = \frac{1}{m+M}\sqrt{\frac{G(\alpha+\gamma)}{\alpha(\alpha-\gamma)}}$$

$$\sigma_2 = \frac{1}{m+M}\sqrt{\frac{G(\alpha-\gamma)}{\alpha(\alpha+\gamma)}}$$

$$\sigma_b = \sqrt{\frac{G}{(m+M)\alpha}}$$

Proof.

$$E = \frac{mM(m+M){\sigma_1}^2}{2} - \frac{GmM}{r_1} = \frac{mM(m+M){\sigma_2}^2}{2} - \frac{GmM}{r_2}$$

$$\frac{mM(m+M)\left({\sigma_1}^2 - {\sigma_2}^2\right)}{2} = -\frac{GmM(r_2 - r_1)}{r_1 r_2}$$

$${\sigma_1}^2 - {\sigma_2}^2 = \frac{4G\gamma}{(m+M)^2(\alpha+\gamma)(\alpha-\gamma)}$$

$$u \coloneqq \frac{\sigma_1}{r_2} = \frac{\sigma_2}{r_1}$$

$$u^2\left((\alpha+\gamma)^2 - (\alpha-\gamma)^2\right) = \frac{4G\gamma}{(m+M)^4(\alpha+\gamma)(\alpha-\gamma)}$$

$$u^2 = \frac{G}{(m+M)^4\alpha(\alpha+\gamma)(\alpha-\gamma)}$$

$$u = \frac{1}{(m+M)^2}\sqrt{\frac{G}{\alpha(\alpha+\gamma)(\alpha-\gamma)}}$$

$$\sigma_1 = \frac{1}{m+M}\sqrt{\frac{G(\alpha+\gamma)}{\alpha(\alpha-\gamma)}}$$

$$\sigma_2 = \frac{1}{m+M}\sqrt{\frac{G(\alpha-\gamma)}{\alpha(\alpha+\gamma)}}$$

$$L = m\frac{Mr_1}{m+M}M\sigma_1 + M\frac{mr_1}{m+M}m\sigma_1$$
$$= mMr_1\sigma_1$$
$$= mMur_1r_2$$
$$= mM\sqrt{\frac{G(\alpha+\gamma)(\alpha-\gamma)}{\alpha}}$$
$$L = mM(m+M)\sqrt{\alpha^2-\gamma^2}\sigma_b$$
$$\sigma_b = \sqrt{\frac{G}{(m+M)\alpha}}$$
$$E = \frac{GmM(\alpha+\gamma)}{2(m+M)\alpha(\alpha-\gamma)} - \frac{GmM}{(m+M)(\alpha-\gamma}$$
$$= \frac{GmM(\alpha+\gamma)-2GmM\alpha}{2(m+M)\alpha(\alpha-\gamma)}$$
$$= -\frac{GmM}{2(m+M)\alpha}$$
$$K = E - U = GmM\left(\frac{1}{r} - \frac{1}{2(m+M)\alpha}\right)$$
$$\sigma = \sqrt{\frac{2K}{mM(m+M)}} = \sqrt{\frac{2G}{m+M}\left(\frac{1}{r} - \frac{1}{2(m+M)\alpha}\right)}$$

□

v 雙星圓軌道互繞

質量 m、M 雙星分別以半徑 $\frac{Mr}{m+M}$、$\frac{mr}{m+M}$ 圓軌道繞雙星質心運動，重力位能 U 以 $r=\infty$ 為 $U=0$，相對雙星質心的慣性參考系中，兩者速率 $M\sigma$、$m\sigma$，角速率 ω，總動能 $K=\frac{mM(m+M)\sigma^2}{2}$，總力學能 $E=U+K$ 稱束縛能，總角動量量值 L：

$$U = -\frac{GmM}{r}$$
$$E = -\frac{GmM}{2r}$$
$$K = \frac{GmM}{2r}$$
$$L = mM\sqrt{\frac{Gr}{m+M}}$$
$$\sigma = \sqrt{\frac{G}{(m+M)r}}$$
$$\omega = \sqrt{\frac{G(m+M)}{r^3}}$$

vi　雙星互相掠過

質量 m、M 雙星自距離無限遠以速率 $(m+M)\eta$ 相互靠近，一星在距離無限遠時過自身位置且以其相對另一星速度為方向向量的直線與另一星距離 b，兩者距離 r，重力位能 U 以 $r=\infty$ 為 $U=0$，相對雙星質心的慣性參考系中，兩者速率 $M\sigma$、$m\sigma$，總動能 $K=\frac{mM(m+M)\sigma^2}{2}$，總力學能 $E=U+K$，總角動量量值 L，雙星最近距離 R，此時兩者速率 $M\zeta$、$m\zeta$：

$$U=-\frac{GMm}{r}$$

$$E=\frac{mM(m+M)\eta^2}{2}$$

$$K=\frac{mM(m+M)\eta^2}{2}+\frac{GMm}{r}$$

$$L=mMb\eta=mMR\zeta$$

$$\sigma=\sqrt{\eta^2+\frac{2G}{(m+M)r}}$$

$$\zeta=\frac{\sqrt{G^2+(m+M)^2\eta^4b^2}+G}{(m+M)\eta b}$$

$$R=\frac{\sqrt{G^2+(m+M)^2\eta^4b^2}-G}{(m+M)\eta^2}$$

Proof.

$$E=\frac{mM(m+M)\sigma^2}{2}-\frac{GMm}{r}=\frac{mM(m+M)\eta^2}{2}$$

$$\sigma^2-\eta^2=\frac{2G}{(m+M)r}$$

$$\sigma=\sqrt{\eta^2+\frac{2G}{(m+M)r}}$$

$$L=m\frac{Mb}{m+M}M\eta+M\frac{mb}{m+M}m\eta=mMb\eta=mMR\zeta$$

$$R=\frac{\eta b}{\zeta}$$

$$\zeta^2-\frac{2G\zeta}{(m+M)\eta b}-\eta^2=0$$

$$\zeta=\frac{G}{(m+M)\eta b}+\sqrt{\left(\frac{G}{(m+M)\eta b}\right)^2+\eta^2}=\frac{G+\sqrt{G^2+(m+M)^2\eta^4b^2}}{(m+M)\eta b}$$

$$R=\frac{(m+M)\eta^2b^2}{G+\sqrt{G^2+(m+M)^2\eta^4b^2}}=\frac{\sqrt{G^2+(m+M)^2\eta^4b^2}-G}{(m+M)\eta^2}$$

□

vii 雙星互繞橢圓軌道的離心率向量（eccentricity vector）、真近點角（True anomaly）、偏近點角（Eccentricity anomaly）與平近點角（Mean anomaly）

離心率向量 $\boldsymbol{\varepsilon}$ 定義為量值等於**離心率** ε、方向為焦點指向近質心點的向量。星體位置 $\mathbf{r}$ 以雙星質心 S 為原點，**真近點角** ν 定義為：

$$\nu = \arccos \frac{\boldsymbol{\varepsilon} \cdot \mathbf{r}}{|\boldsymbol{\varepsilon}||\mathbf{r}|}.$$

半長軸 a，**偏近點角** $\mathscr{E}$ 定義為使得近恆星點為 $\mathscr{E} = 0 + 2z\pi, \quad z \in \mathbb{Z}$ 且逆時針旋轉 $\mathscr{E}$ 增加且：

$$\cos \mathscr{E} = \frac{a - |\mathbf{r}|}{a\varepsilon}.$$

平近點角 $\mathscr{M}$ 定義為：

$$\mathscr{M} = \mathscr{E} - \varepsilon \sin \mathscr{E}.$$

令半短軸 $b = a\sqrt{1-\varepsilon^2}$，以 S 為原點、近質心點在 x 軸正向上，建立平面直角座標，軌道上任一點 P 的位置 (x, y)，則：

$$x = a(\cos \mathscr{E} - \varepsilon)$$

$$y = b \sin \mathscr{E}$$

Proof. 以橢圓中心 C 為圓心、a 為半徑畫輔助圓，將 P 對應到輔助圓上一點 Q 使得直線 PQ 垂直橢圓長軸於 W。建立極座標 $(r; \theta) = \big(r\cos\theta, r\sin\theta\big)$。

$$\cos \angle WCQ = \frac{a\varepsilon + r\cos\theta}{a}$$

$$r = \frac{a(1-\varepsilon^2)}{1 + \varepsilon\cos(\theta)}$$

$$\cos\theta = \frac{a(1-\varepsilon^2) - r}{r\varepsilon}$$

$$\begin{aligned} \cos \angle WCQ &= \frac{a\varepsilon + \frac{a(1-\varepsilon^2)-r}{\varepsilon}}{a} \\ &= \frac{a\varepsilon^2 + a(1-\varepsilon^2) - r}{a\varepsilon} \\ &= \frac{a - r}{a\varepsilon} \end{aligned}$$

故：

$$\angle WCQ = \mathscr{E}$$

$$x = a(\cos \mathscr{E} - \varepsilon)$$

以橢圓中心 C 為圓心、b 為半徑畫輔助圓，將 P 對應到輔助圓上一點 F 使得直線 PF 垂直橢圓短軸於 X，因為：

$$\angle QPF = \angle QWC = \frac{\pi}{2}$$

$$\angle PQF = \angle WQC$$

故：

$$\angle CFX = \angle QFP = \angle WCQ = \mathscr{E}$$

$$y = b\sin\mathscr{E}$$

□

星體與雙星質心連線自近恆星點起掃過的面積 A，週期 T：

$$A = \frac{ab\mathscr{M}}{2}$$

$$\frac{\mathrm{d}\mathscr{M}}{\mathrm{d}t} = \frac{2\pi}{T}$$

Proof.

$$\begin{aligned}
A &= \int_0^{\mathscr{E}} \frac{1}{2}\left(x\frac{\mathrm{d}y}{\mathrm{d}\mathscr{E}} - y\frac{\mathrm{d}x}{\mathrm{d}\mathscr{E}}\right)\mathrm{d}\mathscr{E} \\
&= \int_0^{\mathscr{E}} \frac{1}{2}\left(a(\cos z - \varepsilon)b\cos z + b\sin z a\sin z\right)\mathrm{d}z \\
&= \int_0^{\mathscr{E}} \frac{1}{2}\left(ab - ab\varepsilon\cos z\right)\mathrm{d}z \\
&= \frac{ab}{2}\left(z - \varepsilon\sin z\right)\Big|_0^{\mathscr{E}} \\
&= \frac{ab}{2}\left(\mathscr{E} - \varepsilon\sin\mathscr{E}\right) \\
&= \frac{ab\mathscr{M}}{2}
\end{aligned}$$

□

質量 m、M 雙星分別以橢圓半長軸或圓半徑 a、q 軌道繞雙星質心運動，軌道離心率 ε，週期 T，前者從一個短軸頂點經過近恆星點到達另一個短軸頂點花費時間 T_1，從一個短軸頂點經過遠恆星點到達另一個短軸頂點花費時間 T_2：

$$T = 2\pi\sqrt{\frac{a^3(m+M)^2}{GM^3}} = 2\pi\sqrt{\frac{(a+q)^3}{G(m+M)}}$$

$$T_1 = T\left(\frac{1}{2} - \frac{\varepsilon}{\pi}\right) = \sqrt{\frac{a^3(m+M)^2}{GM^3}}(\pi - 2\varepsilon) = \sqrt{\frac{(a+q)^3}{G(m+M)}}(\pi - 2\varepsilon)$$

$$T_1 = T\left(\frac{1}{2} + \frac{\varepsilon}{\pi}\right) = \sqrt{\frac{a^3(m+M)^2}{GM^3}}(\pi + 2\varepsilon) = \sqrt{\frac{(a+q)^3}{G(m+M)}}(\pi + 2\varepsilon)$$

viii　拉格朗日點（Lagrange points）

雙星以圓軌道互繞，拉格朗日點指使得位於之的一質量極小而可忽略其對雙星重力效應的測試質點可以僅受雙星之重力而維持其與雙星的相對位置的點。令雙星質量 $M > m$，兩者距離 r，則拉格朗日點共有五個，分別是：

- 位於雙星連線段上與質量 m 之星距離 x、與質量 M 之星距離 $r-x$ 的 L_1：

$$\frac{M}{(r-x)^2}-\frac{m}{x^2}=\left(\frac{Mr}{m+M}-x\right)\frac{m+M}{r^3}.$$

- 位於過雙星直線上與質量 m 之星距離 x、與質量 M 之星距離 $r+x$ 的 L_2：

$$\frac{M}{(r+x)^2}+\frac{m}{x^2}=\left(\frac{Mr}{m+M}+x\right)\frac{m+M}{r^3}.$$

- 位於過雙星直線上與質量 m 之星距離 x、與質量 M 之星距離 $x-r$ 的 L_3：

$$\frac{M}{(x-r)^2}+\frac{m}{x^2}=\left(x-\frac{Mr}{m+M}\right)\frac{m+M}{r^3}.$$

- 雙星旋轉平面上距雙星均為 r 的 L_4 與 L_5。

Proof. 雙星系統繞質心角頻率 $\sqrt{\frac{G(M+m)}{r^3}}$，在旋轉平面上建立旋轉參考系，以雙星質心為原點、質量 M 星在 $\left(\frac{m}{M+m}r,0\right)$、質量 m 星在 $\left(-\frac{M}{M+m}r,0\right)$，令 $x_M=\frac{m}{M+m}r$、$x_m=-\frac{M}{M+m}r$、測試質點質量 n。

位能：

$$U(x,y)=-\frac{GMn}{\sqrt{\left(x-x_M\right)^2+y^2}}-\frac{Gmn}{\sqrt{\left(x-x_m\right)^2+y^2}}-\frac{G(M+m)n(x^2+y^2)}{2r^3}.$$

拉格朗日點為 $-\nabla U(x,y)=0$ 處，即：

$$GMn\left(x-x_M\right)\left(\left(x-x_M\right)^2+y^2\right)^{-\frac{3}{2}}+Gmn\left(x-x_m\right)\left(\left(x-x_m\right)^2+y^2\right)^{-\frac{3}{2}}=\frac{G(M+m)nx}{r^3}.$$

$$GMny\left(\left(x-x_M\right)^2+y^2\right)^{-\frac{3}{2}}+Gmny\left(\left(x-x_m\right)^2+y^2\right)^{-\frac{3}{2}}=\frac{G(M+m)ny}{r^3}.$$

即：

$$M\left(x-x_M\right)\left(\left(x-x_M\right)^2+y^2\right)^{-\frac{3}{2}}+m\left(x-x_m\right)\left(\left(x-x_m\right)^2+y^2\right)^{-\frac{3}{2}}=\frac{(M+m)x}{r^3}.$$

$$My\left(\left(x-x_M\right)^2+y^2\right)^{-\frac{3}{2}}+my\left(\left(x-x_m\right)^2+y^2\right)^{-\frac{3}{2}}=\frac{(M+m)y}{r^3}.$$

Case $y=0$：

$$\frac{M}{\left(x-x_M\right)^2}\pm\frac{m}{\left(x-x_m\right)^2}=\frac{M+m}{r^3}|x|.$$

有三個解，分別為 $-x_m<x<x_M$ 的 L_1、$x<-x_m$ 的 L_2 與 $x>x_M$ 的 L_3。

Case $y\neq 0$：

$$M\left(x-x_M\right)\left(\left(x-x_M\right)^2+y^2\right)^{-\frac{3}{2}}+m\left(x-x_m\right)\left(\left(x-x_m\right)^2+y^2\right)^{-\frac{3}{2}}=\frac{(M+m)x}{r^3}.$$

$$M\left(\left(x-x_M\right)^2+y^2\right)^{-\frac{3}{2}}+m\left(\left(x-x_m\right)^2+y^2\right)^{-\frac{3}{2}}=\frac{M+m}{r^3}.$$

$$M\left(x-x_M\right)\left(\left(x-x_M\right)^2+y^2\right)^{-\frac{3}{2}}+m\left(x-x_m\right)\left(\left(x-x_m\right)^2+y^2\right)^{-\frac{3}{2}}=Mx\left(\left(x-x_M\right)^2+y^2\right)^{-\frac{3}{2}}+mx\left(\left(x-x\right.\right.$$

$$-Mx_M\left(\left(x-x_M\right)^2+y^2\right)^{-\frac{3}{2}}-mx_m\left(\left(x-x_m\right)^2+y^2\right)^{-\frac{3}{2}}=0.$$

$$-\frac{Mmr}{M+m}\left(\left(x-x_M\right)^2+y^2\right)^{-\frac{3}{2}}+\frac{Mmr}{M+m}\left(\left(x-x_m\right)^2+y^2\right)^{-\frac{3}{2}}=0.$$

$$x=\frac{x_M+x_m}{2}.$$

$$y=\pm\frac{\sqrt{3}r}{2}.$$

□

XVII　多同質量星正多邊形排列圓軌道互繞

n 個質量均為 m 之星在同一平面上排成一正 n 邊形以半徑 r 圓軌道繞 n 星質心運動，重力位能 U 以任二星之間距離 ∞ 為 $U=0$，相對 n 星質心的慣性參考系中，一星之速率 v，角速率 ω，總動能 $K=\frac{nmv^2}{2}$，總力學能 $E=U+K$ 稱束縛能，總角動量量值 L：

$$U=-\frac{nGm^2}{4r}\sum_{k=1}^{n-1}\csc\frac{\pi k}{n}$$

$$E=-\frac{nGm^2}{8r}\sum_{k=1}^{n-1}\csc\frac{\pi k}{n}$$

$$K=\frac{nGm^2}{8r}\sum_{k=1}^{n-1}\csc\frac{\pi k}{n}$$

$$L=nm\sqrt{\frac{Gmr}{4}\sum_{k=1}^{n-1}\csc\frac{\pi k}{n}}$$

$$v=\sqrt{\frac{Gm}{4r}\sum_{k=1}^{n-1}\csc\frac{\pi k}{n}}$$

$$\omega=\sqrt{\frac{Gm}{4r^3}\sum_{k=1}^{n-1}\csc\frac{\pi k}{n}}$$

Proof.

$$U=-\frac{n}{2}\sum_{k=1}^{n-1}\frac{Gm^2}{\sqrt{2r^2\left(1-\cos\frac{2\pi k}{n}\right)}}$$

$$=-\frac{nGm^2}{4r}\sum_{k=1}^{n-1}\csc\frac{\pi k}{n}$$

$$\omega^2 r = \sum_{k=1}^{n-1} \frac{Gm}{2r\sqrt{2r^2\left(1-\cos\frac{2\pi k}{n}\right)}}$$

$$= \frac{Gm}{4r^2}\sum_{k=1}^{n-1}\csc\frac{\pi k}{n}$$

$$\omega = \sqrt{\frac{Gm}{4r^3}\sum_{k=1}^{n-1}\csc\frac{\pi k}{n}}$$

$$v = \sqrt{\frac{Gm}{4r}\sum_{k=1}^{n-1}\csc\frac{\pi k}{n}}$$

$$K = \frac{nGm^2}{8r}\sum_{k=1}^{n-1}\csc\frac{\pi k}{n}$$

$$L = nmvr = nm\sqrt{\frac{Gmr}{4}\sum_{k=1}^{n-1}\csc\frac{\pi k}{n}}$$

□

XVIII　宇宙膨脹例題

根據霹靂說（The Big Bang Theory），現今宇宙中的所有物質最初是縮聚在一點，在某一瞬間（定義為時間 $t=0$）突然爆炸開來，往各方向均勻膨脹。經許多年後，形成現在的宇宙，且仍然在膨脹中。宇宙中所見的任一星系（Galaxy），對整個宇宙而言，可近似為一個質點。任何一個星系都可選作為宇宙的中心，其他的星系相對於該星系的速度，皆沿徑向遠離，即對此中心為球形對稱。假設宇宙間的質量分布是均勻的，又各星系之間僅有萬有引力的作用，回答下列各題：

i　運動方程

選取某一星系（例如我們所在的銀河系）為座標原點，考慮徑向座標為 r 的另一星系的運動情形。設在半徑為 r 的球體內所含物質的總質量為 M，萬有引力常數為 G，寫出該星系的運動方程式。

答：假設宇宙間的質量分布是均勻的，所以對於座標原點而言，應為球形對稱。設所考慮的星系質量為 m，則該星系所受萬有引力 **F** 為：

$$\mathbf{F} = -\frac{GMm}{r^2}\hat{r}\ldots\boxed{1}$$

設其加速度為 **a** 其運動方程式即：

$$\mathbf{a} = -\frac{GM}{r^2}\hat{r}\ldots\boxed{2}$$

ii　徑向速度作為徑向座標的函數

假設當星系的徑向座標 $r \to \infty$ 時，該星系的徑向速度 $v \leftarrow 0$，求該星系的徑向速度作為其徑向座標的函數。

答：因一維運動，故將向量均視為純量。將二式積分，左式變為功，可換為動能，令積分常數 C，得：

$$\frac{1}{2}v^2 - \frac{GM}{r} = C \dots \boxed{3}$$

$$\lim_{r\to\infty} v = 0.$$

故 $C = 0$。將三式改寫為：

$$v = \sqrt{\frac{2GM}{r}} \dots \boxed{4}$$

iii　徑向座標作為時間的函數

承上，試求徑向座標 r 作為時間 t 的函數。

答：解四式：

$$\sqrt{r}\dot{r} = \sqrt{2GM}$$

$$\int \sqrt{r}\dot{r}\,\mathrm{d}t = \sqrt{2GM}t$$

$$\int \sqrt{r}\dot{r}\,\mathrm{d}t = r^{3/2} - \int \frac{1}{2}\sqrt{r}\,\mathrm{d}t = \frac{2}{3}r^{3/2}$$

$$\frac{2}{3}r^{3/2} = \sqrt{2GM}t$$

$$r = \left(\frac{9}{2}GMt^2\right)^{1/3} \dots \boxed{5}$$

iv　哈柏定律

美國天文學家哈柏經由觀察得知：$v = Hr$，稱為哈柏定律（Hubble's law），式中 H 為哈柏常數（Hubble's constant）。測得 H 值約為 $0.5 \times 10^{-10}\ \mathrm{yr}^{-1}$，求現在所見宇宙的年齡。

答：將五式帶入四式：

$$v = \left(\frac{4GM}{3t}\right)^{1/3} \dots \boxed{6}$$

將六式除以五式：

$$v = \left(\frac{4GM}{3t}\right)^{1/3} \dots \boxed{6}$$

$$r = \left(\frac{9}{2}GMt^2\right)^{1/3} \dots \boxed{5}$$

$$H = \frac{v}{r} = \frac{2}{3t}$$

$$H = \frac{2}{3}t^{-1} = 0.5 \times 10^{-10}\ \mathrm{yr}^{-1}$$

$$t = 1.33 \times 10^{10}\ \mathrm{yr}$$

XIX　振動（Oscillation）

i　（線性）簡諧運動（(Linear) simple harmonic motion, SHM）

簡諧運動指一個一維週期性平移運動，其中一維位置 x 符合：

$$\ddot{x} = -\omega^2(x - C)$$

其中 ω, C 為常數。

解得：

$$x = A\sin(\omega t + \phi) + C$$

$$\dot{x} = A\omega\cos(\omega t + \phi)$$

$$\ddot{x} = -A\omega^2\sin(\omega t + \phi)$$

其中 A 為常數。

角頻率（angular frequency）ω，頻率（frequency）$\frac{\omega}{2\pi}$，週期（period）$\frac{2\pi}{\omega}$，相位角（phase angle）/相常數（phase constant）ϕ，相位（phase）$\omega t+\phi$，振幅（amplitude）$|A|$，平衡點（equilibrium position）C，端點 $C \pm A$，最大速率/速度振幅（velocity amplitude）$|A|\,\omega$，最大加速度量值/加速度振幅（acceleration amplitude）$|A|\,\omega^2$，$x = C \pm A$ 與 $\ddot{x} = \mp A\omega^2$ 發生於 $\omega t + \phi = \frac{\pm\pi}{2} + 2z\pi, \quad z \in \mathbb{Z}$，$x = C$ 發生於 $\omega t + \phi = z\pi, \quad z \in \mathbb{Z}$，$\dot{x} = A\omega$ 發生於 $\omega t + \phi = 2z\pi, \quad z \in \mathbb{Z}$，$\dot{x} = -A\omega$ 發生於 $\omega t + \phi = (2z+1)\,\pi, \quad z \in \mathbb{Z}$。令一垂直 A 之向量 $\mathbf{N}$，任意實數 a 與 b，則 x 恰與一軌跡圓半徑 $|A|$、圓心 $C + a\mathbf{N}$、角速度 $\omega\frac{A \times \mathbf{N}}{|A \times \mathbf{N}|}$ 的等速率圓周運動之位置在直線 $C + tA, \quad t \in \mathbb{A}$ 之正射影相同。

令該物質量 m，受力 F，此裝置稱（線性）簡諧運動振子（(Linear) simple harmonic motion oscillator），並有與虎克定律相同形式之方程：

$$F = -k(\Delta x),$$

其中 $\Delta x = x - C$ 為相對平衡點 C 的位移，$k = m\omega^2$。

F 為保守力，故可定義位能 U：

$$U(\Delta x) = \frac{k(\Delta x)^2}{2} = \frac{m\omega^2 A^2}{2}\sin^2(\omega t + \phi).$$

又動能 K 為：

$$K = \frac{m\dot{x}^2}{2} = \frac{m\omega^2 A^2}{2}\cos^2(\omega t + \phi).$$

兩者和守恆：

$$K + U = \frac{m\omega^2 A^2}{2}.$$

ii 角簡諧運動（Angular simple harmonic motion, angular SHM）/扭擺（Torsion pendulum）

角簡諧運動指一個一維週期性單軸轉動，其中一維角位置 θ 符合：

$$\ddot{\theta} = -\Omega^2(\theta - C)$$

其中 C 為常數，即平衡點，Ω 為常數，即相對平衡點 C 的角位移 $\Delta\theta = \theta - C$ 對時間的函數此週期函數的角頻率（非物之角頻率）。

其相對平衡點 C 的角位移 $\Delta\theta = \theta - C$ 對時間的函數等於一角頻率 Ω 的簡諧運動的相對平衡點的位移對時間的函數的常數倍。

令該物轉動慣量 I，受力矩 τ，此裝置稱角簡諧運動振子（Angular simple harmonic motion oscillator）或扭擺（Torsion pendulum），並有與虎克定律相同形式之方程：

$$\tau = -\kappa(\Delta\theta),$$

其中 $\kappa = m\Omega^2$ 稱扭轉常數（torsion constant）。

iii 彈簧兩端兩物簡諧運動

一質量不計、力常數 k 之理想彈簧，兩端各黏一物，質量 m、M，兩物縮減質量 $\mu = \dfrac{mM}{m+M}$，無外力下進行簡諧運動，則角頻率為 $\sqrt{\dfrac{k}{\mu}}$，若初始時自彈簧被壓縮或拉長 R 靜止釋放，則力學能為 $\dfrac{kR^2}{2}$，以平衡點為相位角 0，則質量 m 者其相對於平衡點的位移 x：

$$x = \frac{\mu R}{m}\sin\left(\sqrt{\frac{k}{\mu}}t\right)$$

$$\dot{x} = \frac{\mu R}{m}\sqrt{\frac{k}{\mu}}\cos\left(\sqrt{\frac{k}{\mu}}t\right)$$

$$\ddot{x} = -\frac{\mu R}{m}\frac{k}{\mu}\sin\left(\sqrt{\frac{k}{\mu}}t\right)$$

彈簧相對於原長的位移 y：

$$y = R\sin\left(\sqrt{\frac{k}{\mu}}t\right)$$

$$\dot{y} = R\sqrt{\frac{k}{\mu}}\cos\left(\sqrt{\frac{k}{\mu}}t\right)$$

$$\ddot{y} = -R\frac{k}{\mu}\sin\left(\sqrt{\frac{k}{\mu}}t\right)$$

總動能 K：

$$K = \frac{kR^2}{2}\cos^2\left(\sqrt{\frac{k}{\mu}}t\right)$$

彈力位能 U：

$$U = \frac{kR^2}{2}\sin^2\left(\sqrt{\frac{k}{\mu}}t\right)$$

iv 簡單單擺（Simple pendulum）小角度近似為角簡諧運動

均勻重力加速度量值 g 下，一質點以長 L（稱擺長）繩懸掛於一固定點，稱樞軸點（pivot point），自相對於樞軸點角位置 $\theta_0 \gtrapprox 0$（即振幅）釋放，稱一簡單單擺，其角位置 θ 對時間的函數服從：

$$L\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} = g\sin\theta$$

小角度近似 $\sin\theta = \theta$：

$$L\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} = g\theta$$

令初始角位置為 $\theta_0 \sin\phi$：

$$\theta = \theta_0 \sin\left(\sqrt{\frac{g}{L}}t + \phi\right)$$

即角頻率 $\sqrt{\frac{g}{L}}$ 的角簡諧運動。

對於真實單擺，常稱為物理單擺（physical pendulum），均勻重力加速度量值 g 下，令其質量 m、轉動慣量 I、重力相對樞軸點之有效力臂 L，則其角位置 θ 對時間的函數服從：

$$I\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} = mgL\sin\theta$$

小角度近似 $\sin\theta = \theta$：

$$I\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} = mgL\theta$$

令初始角位置為 $\theta_0 \sin\phi$：

$$\theta = \theta_0 \sin\left(\sqrt{\frac{I}{mgL}}t + \phi\right)$$

即角頻率 $\sqrt{\frac{I}{mgL}}$ 的角簡諧運動。

v （線性）阻尼運動（(Linearly) damped motion）

線性阻尼運動指一個一維週期性平移運動，其中位置 x 符合：

$$\ddot{x} = -k(x - C) - b\dot{x}$$

其中 $k > 0$、$b > 0$、C 為常數。

令振子質量 m，外力 F，有：

$$F = -mk(x - C) - mb\dot{x},$$

其中 mb 稱阻尼係數（damping constant）。

- $b^2 > 4k$ 稱**過阻尼（Overdamped）**

$$x = Ae^{\frac{-b}{2}t}\left(e^{\frac{\sqrt{b^2-4k}}{2}t} + Be^{\frac{-\sqrt{b^2-4k}}{2}t}\right) + C$$

- $b^2 = 4k$ 稱**臨界阻尼（Critically damped）**：

$$x = A(1 + Bt)e^{\frac{-b}{2}t} + C$$

- $b^2 < 4k$ 稱**欠阻尼/次阻尼（Underdamped）**：

$$x = Ae^{\frac{-b}{2}t}\sin\left(\frac{\sqrt{4k - b^2}}{2}t + \phi\right) + C$$

vi　均勻有效重力場下鉛直圓周運動或簡單單擺運動

均勻有效重力場量值 g 下，令重力加速度 $(g, 0)$，xy 平面上一圓以原點為圓心、半徑 R，稱軌跡圓，定義角位置 θ 使得圓周上一點位置為 $(R\cos\theta, R\sin\theta)$，一質量 m 質點在圓周上僅受重力與一垂直圓周之外力 $\mathbf{N} = N(-\cos\theta, -\sin\theta)$，$t = 0$ 時恰以速度 $(0, u)$ 通過圓周最低點 $(R, 0)$，其中 $u \geq 0$。當質點在圓周上且以原點為參考點的角速度出紙面時，令速率 v，以原點為參考點的角速度 $\dot{\theta}$，角加速度 $\ddot{\theta}$，切向加速度 $R\ddot{\theta}(-\sin\theta, \cos\theta)$，$a \coloneqq R\ddot{\theta}$，法向加速度 $R\dot{\theta}^2(-\cos\theta, -\sin\theta)$，$q \coloneqq R\dot{\theta}^2$，動能 K，重力位能 U 以 y 軸為 $U = 0$，力學能 $E = U + K$。若此運動順利通過圓周最高點 $(-R, 0)$，則不斷逆時針繞原點不等速率地公轉，稱鉛直圓周運動，令週期 T_1；若此運動不通過最高點而在圓周上兩 θ 互為相反數的點之間週期往復運動，稱簡單單擺運動或單擺運動，令該二點 $\theta = \pm\theta_m$，其中 $\theta_m > 0$，稱臨界角（critical angle），質點在該二點時速率為零，質點每次通過 $(0, R)$ 時之速率均為 u 但與前一次通過時之速度反向，此運動相當於受力條件不變下自 $(R\cos\theta_m, \pm R\sin\theta_m)$ 靜止釋放之運動，令週期（自 $\theta = \theta_m$ 回到 $\theta = \theta_m$ 所須的時間）T_2：

$$U = -mgR\cos\theta$$

$$E = U + \frac{mv^2}{2} = \frac{mu^2}{2} - mgR$$

$$K = \frac{mv^2}{2} = \frac{mu^2}{2} - mgR(1 - \cos\theta)$$

$$v = \sqrt{u^2 - 2gR(1 - \cos\theta)}$$

$$\dot{\theta} = \frac{v}{R} = \frac{\sqrt{u^2 - 2gR(1 - \cos\theta)}}{R}$$

$$q = \frac{v^2}{R} = \frac{u^2 - 2gR(1 - \cos\theta)}{R}$$

$$a = g\sin\theta$$

$$\ddot{\theta} = \frac{a}{R} = \frac{g}{R}\sin\theta$$

$$N = \frac{mv^2}{R} - mg\cos\theta = \frac{m}{R}\left(u^2 - 2gR + gR\cos\theta\right)$$

$$T_1 = \int_0^{2\pi}\frac{\mathrm{d}\theta}{\dot{\theta}} = R\int_0^{2\pi}\frac{\mathrm{d}\theta}{\sqrt{u^2 - 2gR(1 - \cos\theta)}}$$

$$\theta_m = \arccos\frac{u^2}{2gR}$$

$$T_2 = 2\int_{-\theta_m}^{\theta_m} \frac{\mathrm{d}\theta}{\dot{\theta}} = 4R\int_{0}^{\theta_m} \frac{\mathrm{d}\theta}{\sqrt{u^2 - 2gR(1-\cos\theta)}}$$

當限制條件 $N \geq 0$ 時（例如以連接圓心與質點之質量不計軟繩提供之繩張力為 **N**，或以軌跡圓圓周的光滑可脫離軌道提供之正向力為 **N**），則：

- $u > \sqrt{5GR}$：鉛直圓周運動，最高點 $N > 0$、$v > \sqrt{gR}$。
- $u = \sqrt{5GR}$：鉛直圓周運動，最高點 $N = 0$、$v = \sqrt{gR}$。
- $\sqrt{2gR} < u < \sqrt{5gR}$：於 y 軸脫離軌跡圓。
- $0 < u \leq \sqrt{2gR}$：單擺運動。
- $u = 0$：靜止於 $(R, 0)$。

當 N 無限制條件時（例如以連接圓心與質點之質量不計剛體提供之力為 **N**，或以軌跡圓圓周的光滑不可脫離軌道提供之力為 **N**），則：

- $u > \sqrt{5GR}$：鉛直圓周運動，最高點 $N > 0$、$v > \sqrt{gR}$。
- $u = \sqrt{5GR}$：鉛直圓周運動，最高點 $N = 0$、$v = \sqrt{gR}$。
- $2\sqrt{gR} < u < \sqrt{5gR}$：鉛直圓周運動，最高點 $-mg < N < 0$、$0 < v < \sqrt{gR}$。
- $u = 2\sqrt{gR}$：到達最高點時恰停止，最高點 $N = -mg$、$v = 0$。
- $0 < u \leq 2\sqrt{gR}$：單擺運動。
- $u = 0$：靜止於 $(R, 0)$。

vii Forced Oscillations and Resonance

If an external driving force with angular frequency ω_d acts on an oscillating system with natural angular frequency ω, the system oscillates with angular frequency ω_d. The velocity amplitude of the system is greatest when

$$\omega_d = \omega,$$

a condition called resonance. The amplitude of the system is approximately greatest under the same condition.

XX Lagrangian Mechanics (拉格朗日力學)

i Configuration space

The degrees of freedom, or parameters, that define the configuration, or position of all constituent point particles, of a physical system are called generalized coordinates, and the space defined by these coordinates is called the configuration space of the system. Each unique possible configuration of the system corresponds to a unique

point in the space. It is often the case that the set of all actual configurations of the system is a manifold in the configuration space, called the configuration manifold of the system. The number of dimensions of the configuration space is equal to the degrees of freedom of the system.

ii Lagrangian mechanics

Lagrangian mechanics describes a mechanical system as a pair (M, L) consisting of a configuration manifold M and a smooth functional $L : TM \to \mathbb{R}$, called a Lagrangian, where TM is the tagent bundle of M, that is, $L = L(\mathbf{q}, \dot{\mathbf{q}}, t)$, where the vector $\mathbf{q}$ is a point in the configuration space of the system, $\dot{\mathbf{q}} = \dfrac{\mathrm{d}\mathbf{q}}{\mathrm{d}t}$, and t is time.

iii Non-Relativistic Lagrangian

Let $\mathbf{r}_k$ denote the position of the kth particle for $_{k=1}^{N}$ in the position space $\mathbb{R}^3$ equipped with Cartesian coordinates as function of time, and

$$\mathbf{v}_k = \dot{\mathbf{r}_k} = \frac{\mathrm{d}\mathbf{r}_k}{\mathrm{d}t}.$$

The non-relativistic Lagrangian for a system of N particles is given by

$$L = T - V,$$

where

$$T = \frac{1}{2}\sum_{k=1}^{N} m_k {\mathbf{v}_k}^2$$

is the total kinetic energy of the system. Each particle labeled k has mass m, and ${\mathbf{v}_k}^2 = \mathbf{v}_k \cdot \mathbf{v}_k$ is the square of the norm of its velocity, equivalent to the dot product of the velocity with itself.

V, the potential energy of the system, is the energy any one of the particles has due to all the others together with all external influences. For conservative forces (e.g. Newtonian gravity), V is a function of the vectors of the position of the particles only, that is, $V = V((\mathbf{r}_k)_{i=1}^{N})$. For non-conservative forces that can be derived from an appropriate potential (e.g. electromagnetic potential), V is a function of the vectors of the position and velocity of the particles, that is, $V = V((\mathbf{r}_k)_{i=1}^{N}, (\mathbf{v}_k)_{i=1}^{N})$. For some time-dependent external field or force (e.g. electric field and magnetic flux density field in electromagnetodynamics), the potential changes with time, so most generally, V is a function of the vectors of the position and velocity of the particles and time, that is, $V = V((\mathbf{r}_k)_{i=1}^{N}, (\mathbf{v}_k)_{i=1}^{N}, t)$.

iv Lagrange's Equations of the First Kind

With these definitions, Lagrange's equations of the first kind are

$$\nabla_{\mathbf{r}_k} L - \frac{\mathrm{d}}{\mathrm{d}t} \nabla_{\dot{\mathbf{r}}_k} L + \sum_{i=1}^{C} \lambda_i \nabla_{\mathbf{r}_k} f_i = 0,$$

where k for $\overset{N}{k=1}$ labels the particles, f_i for $\overset{C}{i=1}$ labels C constraint equations, and λ_i for $\overset{C}{i=1}$ labels the Lagrange multiplier for the ith constraint equation. The constraint equations can be either holonomic, that is, in the form of $f_i((\mathbf{r}_k)_{i=1}^N) = 0$, non-holonomic, that is, in the form of $f_i((\mathbf{r}_k)_{i=1}^N, (\mathbf{v}_k)_{i=1}^N) = 0$, or most generally, time dependent, that is, in the form of $f_i((\mathbf{r}_k)_{i=1}^N, (\mathbf{v}_k)_{i=1}^N, t) = 0$.

v From position space to configuration space

In each constraint equation, one coordinate can be determined from the other coordinates. The number of independent coordinates is therefore $n = 3N - C$. We can construct a configuration space with n generalized coordinates, conveniently written as an n-vector $\mathbf{q} = ((q_k)_{k=1}^n)$. Hence the position coordinates as functions of the generalized coordinates and time are

$$\mathbf{r}_k : \mathbb{R}^{n+1} \to \mathbb{R}^3;\ (\mathbf{q}, t) \mapsto \mathbf{r}_k(x_k, y_k, z_k).$$

The vector $\mathbf{q}$ is a point in the configuration space of the system. The time derivatives of the generalized coordinates are called the generalized velocities, and for each particle labelled as k for $\overset{N}{k=1}$, its velocity vector, namely, the total derivative of its position with respect to time, is

$$\mathbf{v}_k = \left(\dot{\mathbf{q}} \cdot \nabla_{\mathbf{q}}\right) \mathbf{r}_k + \frac{\partial \mathbf{r}_k}{\partial t}.$$

vi Euler–Lagrange Equations or Lagrange's Equations of the Second Kind

With these definitions, the Euler–Lagrange equations, or Lagrange's equations of the second kind are

$$\nabla_{\mathbf{q}} L = \frac{\mathrm{d}}{\mathrm{d}t} \nabla_{\dot{\mathbf{q}}} L.$$

The number of equations has decreased compared to Newtonian mechanics, from $3N$ to $n = 3N - C$ coupled second-order differential equations in the generalized coordinates. These equations do not include constraint forces at all, only non-constraint forces need to be accounted for.